Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: KISST DAIRY #1

Physical address of dairy:

Number and Street	City	County	Zip Code
20000 CEDAR AVE	TRACY	San Joaquin	95304

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1992

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X213-X190-XX09-XXXX X213-X190-XX16-XXXX X213-X190-XX17-XXXX

B. OPERATORS

Kisst, John

Operator name: Kisst, John

Telephone no.: Landline: ; Cellular: (209) 481-1188

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 300	Ripon	CA	95366

This operator is responsible for paying permit fees.

C. OWNERS

Kisst, John

Legal owner name: Kisst, John

Telephone no.: Landline: ; Cellular: (209) 481-1188

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 300	Ripon	CA	95366

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	900	800	300	0
Number under roof	1,000	70	0	0	0	0
Maximum number	1,000	70	900	800	300	0
Average number	1,000	70	900	800	300	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 45,877.91 *tons per reporting period*

Total nitrogen from manure: 540,159.70 *lbs per reporting period*

After ammonia losses (30% loss applied): 378,111.79 *lbs per reporting period*

Total phosphorus from manure: 89,671.63 *lbs per reporting period*

Total potassium from manure: 198,079.23 *lbs per reporting period*

Total salt from manure: 486,946.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 20,160,000 *gallons*

Total nitrogen generated: 96,731.48 *lbs*

Total phosphorus generated: 3,955.76 *lbs*

Total potassium generated: 307,126.88 *lbs*

Total salt generated: 1,365,674.27 *lbs*

	20,160,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	20,160,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal SSJID	Surface water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/07/2023	Solid commercial fertilizer *UN32*	41.00 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/18/2023	Solid commercial fertilizer *4-10-10*	35.60 *ton*	Dry-weight	0.1	4.000000	10.000000	10.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	29,058.91	7,112.88	7,112.88	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	29,058.91	7,112.88	7,112.88	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
C-1	69	69	2	manure	X213-X150-XX08-XXXX
C-2	74	74	2	manure	X213-X150-XX22-XXXX
CC-1	421	421	1	none	X008-X340-X035-XXXX
CC-2	250	250	1	manure	X008-X340-X042-XXXX
CC-3	215	215	1	none	X002-X250-X001-XXXX
CC-4	318	318	1	none	X002-X250-X003-XXXX X002-X250-X006-XXXX
CC-5	277	277	1	none	X002-X230-X001-XXXX
CC-6	253	253	1	none	X002-X250-X004-XXXX
CC-7	274	274	1	none	X002-X220-X001-XXXX X002-X230-X001-XXXX
CC-8	136	136	1	none	X002-X240-X001-XXXX
CC-B2	64	64	1	none	X002-X250-X004-XXXX
CC-B3	51	51	1	none	X002-X250-X004-XXXX
CC-D5	42	42	1	none	X008-X340-X035-XXXX
CC-P8	28	28	1	none	X002-X240-X001-XXXX
CC-W1	52	52	2	manure	X002-X250-X001-XXXX
CC-W2	55	55	2	manure	X002-X250-X001-XXXX
Escalon	45	45	1	manure	X203-X200-XX01-XXXX X203-X200-XX02-XXXX X203-X200-XX22-XXXX
K1-1	35	35	2	process wastewater	X213-X260-XX23-XXXX
K1-10	84	84	1	process wastewater	X213-X190-XX02-XXXX
K1-2	40	40	2	process wastewater	X213-X260-XX23-XXXX
K1-3	12	12	2	process wastewater	X213-X260-XX23-XXXX
K1-4	11	11	0	none	X213-X190-XX16-XXXX
K1-5	12	12	2	process wastewater	X213-X190-XX09-XXXX
K1-6	20	20	2	process wastewater	X213-X190-XX09-XXXX
K1-7	40	40	2	process wastewater	X213-X190-XX09-XXXX
K1-8	34	34	2	process wastewater	X213-X260-XX05-XXXX
K1-9	40	40	1	manure	X213-X260-XX27-XXXX
Totals for areas that were used for application	862	862	26		
Totals for areas that were not used for application	2,090	2,090	11		
Land application area totals	2,952	2,952	37		

B. CROPS AND HARVESTS

C-1

Field name: C-1

12/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 69 Plant date: 12/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	932.00 *ton*	As-is		61.1	5,610.00	1,100.00	4,740.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.51	151.55	29.72	128.05	1,092.90

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 69 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,942.00 *ton*	As-is		69.6	4,040.00	560.00	5,150.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	28.14	227.41	31.52	289.89	1,129.40

C-2

Field name: C-2

C-2

12/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 74 Plant date: 12/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	1,102.00 *ton*	As-is		55.1	4,780.00	1,140.00	4,950.00		9.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	14.89	142.37	33.95	147.43	1,305.20

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 74 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,974.00 *ton*	As-is		67.4	4,240.00	840.00	6,530.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.68	226.21	44.82	348.38	1,043.55

CC-1

Field name: CC-1

11/17/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 421 Plant date: 11/17/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	3,021.00 *ton*	As-is		12.4	26,500.00	1,600.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.18	380.32	22.96	119.12	1,206.91

CC-2

Field name: CC-2

11/09/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 250 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,684.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.74	357.01	21.02	111.82	1,132.94

CC-3

Field name: CC-3

11/12/2017: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 215 Plant date: 11/12/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,422.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.61	350.54	20.64	109.79	1,112.41

CC-4

Field name: CC-4

CC-4

11/06/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 318 Plant date: 11/06/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	2,645.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	8.32	440.83	25.95	138.07	1,398.96

CC-5

Field name: CC-5

11/06/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 277 Plant date: 11/06/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,934.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.98	370.04	21.78	115.90	1,174.31

CC-6

Field name: CC-6

CC-6

11/22/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 253 Plant date: 11/22/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,812.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.16	379.59	22.35	118.89	1,204.60

CC-7

Field name: CC-7

11/22/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 274 Plant date: 11/22/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,742.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.36	336.96	19.84	105.54	1,069.31

CC-8

Field name: CC-8

CC-8

11/23/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 136 Plant date: 11/23/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	862.00 *ton*	As-is		12.4	26,500.00	1,560.00	8,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.34	335.93	19.78	105.21	1,066.04

CC-B2

Field name: CC-B2

10/24/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 64 Plant date: 10/24/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	423.00 *ton*	As-is		13.6	28,800.00	1,500.00	7,600.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.61	380.70	19.83	100.46	982.21

CC-B3

Field name: CC-B3

CC-B3

10/24/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 51 Plant date: 10/24/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	306.00 *ton*	As-is		13.6	28,800.00	1,520.00	7,580.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.00	345.60	18.24	90.96	891.65

CC-D5

Field name: CC-D5

10/25/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 42 Plant date: 10/25/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	245.00 *ton*	As-is		13.6	28,800.00	1,500.00	7,600.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	5.83	336.00	17.50	88.67	866.88

CC-P8

Field name: CC-P8

CC-P8

10/25/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 28 Plant date: 10/25/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	198.00 *ton*	As-is		13.6	28,800.00	1,520.00	7,580.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.07	407.31	21.50	107.20	1,050.87

CC-W1

Field name: CC-W1

12/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 12/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	703.00 *ton*	As-is		62.4	5,000.00	940.00	5,940.00		9.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.52	135.19	25.42	160.61	1,002.41

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,158.00 *ton*	As-is		68.0	3,840.00	550.00	4,660.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	22.27	171.03	24.50	207.55	783.88

CC-W2

Field name: CC-W2

12/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 12/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	722.00 *ton*	As-is		60.4	5,490.00	1,070.00	6,930.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.13	144.14	28.09	181.94	1,133.25

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,263.00 *ton*	As-is		66.4	4,070.00	550.00	4,620.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	160.00	30.00	132.00	0.00
Total actual harvest content	22.96	186.92	25.26	212.18	879.60

Escalon

Field name: Escalon

02/01/2012: Almond, in shell

Crop: Almond, in shell Acres planted: 45 Plant date: 02/01/2012

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	303.00 *ton*	As-is		7.4	11,800.00	1,740.00	19,600.00		15.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	6.73	158.91	23.43	263.95	1,920.40

K1-1

Field name: K1-1

12/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 12/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	396.00 *ton*	As-is		51.8	6,690.00	1,130.00	5,170.00		7.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	11.31	151.39	25.57	116.99	871.47

09/02/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 09/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/13/2023	312.00 *ton*	As-is		68.2	5,680.00	1,030.00	9,150.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.91	101.27	18.36	163.13	793.73

K1-10

Field name: K1-10

03/13/2021: Oats, hay

Crop: Oats, hay Acres planted: 84 Plant date: 03/13/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	598.00 *ton*	As-is		14.0	16,300.00	3,030.00	18,600.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	320.00	52.00	264.00	0.00
Total actual harvest content	7.12	232.08	43.14	264.83	1,199.99

K1-2

Field name: K1-2

12/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 12/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	485.00 *ton*	As-is		55.6	6,030.00	1,120.00	5,020.00		9.84

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.13	146.23	27.16	121.74	1,059.47

09/02/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 40 Plant date: 09/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/13/2023	379.00 *ton*	As-is		67.0	4,650.00	1,020.00	9,950.00		17.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.48	88.12	19.33	188.55	1,069.35

K1-3

Field name: K1-3

12/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 12/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	157.00 *ton*	As-is		55.6	6,500.00	1,070.00	4,470.00		9.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.08	170.08	28.00	116.97	1,076.99

K1-3

09/02/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 12 Plant date: 09/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/13/2023	83.00 *ton*	As-is		66.8	7,450.00	1,120.00	10,400.00		14.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	6.92	103.06	15.49	143.87	656.75

K1-5

Field name: K1-5

12/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 12/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	168.00 *ton*	As-is		60.0	5,950.00	1,020.00	4,890.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	14.00	166.60	28.56	136.92	1,198.40

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	312.00 *ton*	As-is		69.4	4,070.00	630.00	5,760.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.00	211.64	32.76	299.52	1,129.75

K1-6

Field name: K1-6

12/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 12/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	288.00 *ton*	As-is		60.4	5,340.00	990.00	4,630.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	14.40	153.79	28.51	133.34	1,208.91

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	512.00 *ton*	As-is		68.6	4,270.00	580.00	4,760.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.60	218.62	29.70	243.71	1,077.15

K1-7

Field name: K1-7

12/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 12/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	601.00 *ton*	As-is		59.4	5,210.00	1,110.00	5,000.00		9.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	15.03	156.56	33.36	150.25	1,156.59

K1-7

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	1,049.00 *ton*	As-is		66.4	4,070.00	550.00	4,620.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.23	213.47	28.85	242.32	934.03

K1-8

Field name: K1-8

12/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 12/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	402.00 *ton*	As-is		55.9	6,480.00	1,190.00	5,900.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	11.82	153.23	28.14	139.52	1,063.69

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	936.00 *ton*	As-is		70.2	3,960.00	600.00	5,970.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.53	218.03	33.04	328.70	1,115.71

K1-9

Field name: K1-9

11/17/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	289.00 *ton*	As-is		13.6	28,800.00	1,520.00	7,580.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.23	416.16	21.96	109.53	1,073.69

NUTRIENT BUDGET

A. LAND APPLICATIONS

C-1 - 12/23/2022: Oats, silage-soft dough

Field name: C-1

Crop: Oats, silage-soft dough — Plant date: 12/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	198.59	107.40	200.18	4,114.87	710.00 *ton*
Application event totals		198.59	107.40	200.18	4,114.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.95	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.95	

C-1 - 06/17/2023: Corn, silage

Field name: C-1

Crop: Corn, silage — Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	234.90	72.38	348.41	5,495.93	750.00 *ton*
Application event totals		234.90	72.38	348.41	5,495.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	66.76	11,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.76	

C-1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	65.31	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.31	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	65.31	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.31	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.58	10,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.58	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	63.86	10,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.86	

C-1 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	63.13	10,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.13	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	62.41	10,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.41	

C-2 - 12/23/2022: Oats, silage-soft dough

Field name: C-2

Crop: Oats, silage-soft dough Plant date: 12/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	193.00	104.37	194.54	3,998.96	740.00 *ton*
Application event totals		193.00	104.37	194.54	3,998.96	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.19	10,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.19	

C-2 - 06/17/2023: Corn, silage

Field name: C-2

Crop: Corn, silage Plant date: 06/17/2023

C-2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	230.71	71.09	342.20	5,397.90	790.00 *ton*
Application event totals		230.71	71.09	342.20	5,397.90	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.54	10,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.54	

06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.19	10,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.19	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.51	10,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.51	

C-2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.84	10,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.48	9,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.13	9,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.13	9,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

CC-1 - 11/17/2021: Alfalfa, hay

Field name: CC-1

Crop: Alfalfa, hay

Plant date: 11/17/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	50.19	50,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.19	

CC-1 - 11/17/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.48	49,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.48	49,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.24	49,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	48.64	49,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	48.29	48,720,000.00 *gal*
River	Surface water	0.00	0.00	0.00	48.29	48,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	96.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	47.93	48,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.93	

CC-2 - 11/09/2022: Alfalfa, hay

Field name: CC-2

Crop: Alfalfa, hay

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	266.34	144.04	268.47	5,518.56	3,450.00 *ton*
Application event totals		266.34	144.04	268.47	5,518.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.08	35,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.28	34,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.28	34,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.08	34,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.68	34,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.68	

CC-2 - 11/09/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.88	34,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.88	

09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.28	34,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.28	

CC-3 - 11/12/2017: Alfalfa, hay

Field name: CC-3

Crop: Alfalfa, hay

Plant date: 11/12/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	61.95	31,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.95	

05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	26.32	13,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.32	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.39	30,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.39	

CC-3 - 11/12/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.06	29,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.45	30,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.82	29,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.89	28,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.89	

CC-4 - 11/06/2018: Alfalfa, hay

Field name: CC-4

Crop: Alfalfa, hay

Plant date: 11/06/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	50.07	38,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.07	

CC-4 - 11/06/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.28	37,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.28	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.44	37,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.44	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	48.18	36,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.18	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	48.34	36,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.34	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	48.02	36,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.02	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	49.44	37,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.44	

CC-5 - 11/06/2018: Alfalfa, hay

CC-5 - 11/06/2018: Alfalfa, hay

Field name: CC-5

Crop: Alfalfa, hay

Plant date: 11/06/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.20	38,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.48	38,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.12	37,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.76	37,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.85	37,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.31	36,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.31	

CC-5 - 11/06/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.77	36,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.77	

CC-6 - 11/22/2019: Alfalfa, hay

Field name: CC-6

Crop: Alfalfa, hay

Plant date: 11/22/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.16	36,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.16	

04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.17	35,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.17	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.19	34,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.19	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.18	35,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.18	

CC-6 - 11/22/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.80	34,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.81	33,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.22	33,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.22	

CC-7 - 11/22/2019: Alfalfa, hay

Field name: CC-7

Crop: Alfalfa, hay

Plant date: 11/22/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	53.54	35,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	52.45	34,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.45	

CC-7 - 11/22/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	51.17	33,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	51.53	33,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	51.53	33,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	50.62	33,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	50.25	33,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.25	

CC-8 - 11/23/2019: Alfalfa, hay

Field name: CC-8

Crop: Alfalfa, hay

Plant date: 11/23/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

CC-8 - 11/23/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	62.96	20,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	61.11	19,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.75	19,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.01	19,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.27	19,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.70	18,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.70	

CC-8 - 11/23/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.17	18,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.17	

CC-B2 - 10/24/2020: Alfalfa, hay

Field name: CC-B2

Crop: Alfalfa, hay

Plant date: 10/24/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	61.81	9,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.46	9,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.46	9,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.68	9,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.68	

CC-B2 - 10/24/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.89	8,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.89	8,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.11	8,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.11	

CC-B3 - 10/24/2020: Alfalfa, hay

Field name: CC-B3

Crop: Alfalfa, hay

Plant date: 10/24/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.80	7,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	59.89	7,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.89	

CC-B3 - 10/24/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.87	7,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.91	7,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.91	7,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.94	6,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.98	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.98	

CC-D5 - 10/25/2020: Alfalfa, hay

Field name: CC-D5

Crop: Alfalfa, hay

Plant date: 10/25/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

CC-D5 - 10/25/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	69.14	6,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	69.14	6,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	67.95	6,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	66.76	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	65.57	6,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.38	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.38	

CC-D5 - 10/25/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	63.18	6,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.18	

CC-P8 - 10/25/2020: Alfalfa, hay

Field name: CC-P8

Crop: Alfalfa, hay

Plant date: 10/25/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	73.32	4,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	73.32	4,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	71.53	4,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	71.53	4,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.53	

CC-P8 - 10/25/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	67.95	4,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.38	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.38	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.38	

CC-W1 - 12/27/2022: Oats, silage-soft dough

Field name: CC-W1

Crop: Oats, silage-soft dough Plant date: 12/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	174.44	94.34	175.84	3,614.44	470.00 *ton*
Application event totals		174.44	94.34	175.84	3,614.44	

CC-W1 - 06/18/2023: Corn, silage

Field name: CC-W1

Crop: Corn, silage Plant date: 06/18/2023

CC-W1 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	149.61	46.10	221.91	3,500.49	360.00 *ton*
Application event totals		149.61	46.10	221.91	3,500.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.74	7,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.77	7,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.85	6,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.85	

CC-W1 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	53.92	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	52.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	51.03	6,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.03	

CC-W2 - 12/27/2022: Oats, silage-soft dough

Field name: CC-W2

Crop: Oats, silage-soft dough

Plant date: 12/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	175.45	94.89	176.86	3,635.42	500.00 *ton*
Application event totals		175.45	94.89	176.86	3,635.42	

CC-W2 - 06/18/2023: Corn, silage

Field name: CC-W2

Crop: Corn, silage

Plant date: 06/18/2023

CC-W2 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	161.10	49.64	238.95	3,769.21	410.00 *ton*
Application event totals		161.10	49.64	238.95	3,769.21	

06/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

06/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	66.46	8,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.46	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	64.64	8,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.64	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	63.73	8,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.73	

CC-W2 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	62.82	8,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	61.90	8,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	61.90	8,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.90	

Escalon - 02/01/2012: Almond, in shell

Field name: Escalon

Crop: Almond, in shell

Plant date: 02/01/2012

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	98.64	53.35	99.43	2,043.91	230.00 *ton*
Application event totals		98.64	53.35	99.43	2,043.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.21	6,062,659.00 *gal*
Application event totals		0.00	0.00	0.00	56.21	

Escalon - 02/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	112.43	12,125,318.00 *gal*
Application event totals		0.00	0.00	0.00	112.43	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	112.43	12,125,318.00 *gal*
Application event totals		0.00	0.00	0.00	112.43	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	112.43	12,125,318.00 *gal*
Application event totals		0.00	0.00	0.00	112.43	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	112.43	12,125,318.00 *gal*
Application event totals		0.00	0.00	0.00	112.43	

Escalon - 02/01/2012: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.21	6,062,659.00 *gal*
Application event totals		0.00	0.00	0.00	56.21	

K1-1 - 12/21/2022: Oats, silage-soft dough

Field name: K1-1

Crop: Oats, silage-soft dough

Plant date: 12/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	132.61	6.23	460.93	1,910.67	1,080,000.00 *gal*
Application event totals		132.61	6.23	460.93	1,910.67	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.64	2.29	238.91	1,135.87	600,000.00 *gal*
River	Surface water	0.00	0.00	0.00	68.67	5,760,000.00 *gal*
Application event totals		49.64	2.29	238.91	1,204.54	

K1-1 - 09/02/2023: Sudangrass, silage

Field name: K1-1

Crop: Sudangrass, silage

Plant date: 09/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

K1-1 - 09/02/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	71.53	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.35	2.94	302.14	1,396.24	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	67.24	5,640,000.00 *gal*
Application event totals		65.35	2.94	302.14	1,463.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.35	2.94	302.14	1,396.24	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	62.95	5,280,000.00 *gal*
Application event totals		65.35	2.94	302.14	1,459.18	

K1-10 - 03/13/2021: Oats, hay

Field name: K1-10

Crop: Oats, hay

Plant date: 03/13/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.64	2.29	238.91	1,135.87	1,440,000.00 *gal*
River	Surface water	0.00	0.00	0.00	58.42	11,760,000.00 *gal*
Application event totals		49.64	2.29	238.91	1,194.29	

K1-10 - 03/13/2021: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.64	2.29	238.91	1,135.87	1,440,000.00 *gal*
River	Surface water	0.00	0.00	0.00	57.22	11,520,000.00 *gal*
Application event totals		49.64	2.29	238.91	1,193.10	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.43	11,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.43	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.22	11,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.22	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	56.63	11,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.63	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.84	11,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.84	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	164.52	5.18	244.63	1,038.59	1,440,000.00 *gal*
River	Surface water	0.00	0.00	0.00	54.84	11,040,000.00 *gal*
Application event totals		164.52	5.18	244.63	1,093.43	

K1-10 - 03/13/2021: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	54.24	10,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.24	

K1-2 - 12/21/2022: Oats, silage-soft dough

Field name: K1-2

Crop: Oats, silage-soft dough

Plant date: 12/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.82	6.66	492.94	2,043.36	1,320,000.00 *gal*
Application event totals		141.82	6.66	492.94	2,043.36	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.12	2.40	250.85	1,192.67	720,000.00 *gal*
River	Surface water	0.00	0.00	0.00	66.34	6,360,000.00 *gal*
Application event totals		52.12	2.40	250.85	1,259.01	

K1-2 - 09/02/2023: Sudangrass, silage

Field name: K1-2

Crop: Sudangrass, silage

Plant date: 09/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

K1-2 - 09/02/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.08	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.18	2.57	264.37	1,221.71	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	57.58	5,520,000.00 *gal*
Application event totals		57.18	2.57	264.37	1,279.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.18	2.57	264.37	1,221.71	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	56.33	5,400,000.00 *gal*
Application event totals		57.18	2.57	264.37	1,278.04	

K1-3 - 12/21/2022: Oats, silage-soft dough

Field name: K1-3

Crop: Oats, silage-soft dough

Plant date: 12/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.91	8.08	597.50	2,476.80	480,000.00 *gal*
Application event totals		171.91	8.08	597.50	2,476.80	

K1-3 - 12/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.44	2.00	209.04	993.89	180,000.00 *gal*
River	Surface water	0.00	0.00	0.00	62.59	1,800,000.00 *gal*
Application event totals		43.44	2.00	209.04	1,056.48	

K1-3 - 09/02/2023: Sudangrass, silage

Field name: K1-3

Crop: Sudangrass, silage

Plant date: 09/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	66.76	1,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.47	3.22	330.46	1,527.14	180,000.00 *gal*
River	Surface water	0.00	0.00	0.00	58.42	1,680,000.00 *gal*
Application event totals		71.47	3.22	330.46	1,585.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.56	2.68	275.39	1,272.61	150,000.00 *gal*
River	Surface water	0.00	0.00	0.00	54.24	1,560,000.00 *gal*
Application event totals		59.56	2.68	275.39	1,326.86	

K1-5 - 12/22/2022: Oats, silage-soft dough

K1-5 - 12/22/2022: Oats, silage-soft dough

Field name: K1-5

Crop: Oats, silage-soft dough

Plant date: 12/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.91	8.08	597.50	2,476.80	480,000.00 *gal*
Application event totals		171.91	8.08	597.50	2,476.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.44	2.00	209.04	993.89	180,000.00 *gal*
River	Surface water	0.00	0.00	0.00	64.67	1,860,000.00 *gal*
Application event totals		43.44	2.00	209.04	1,058.56	

K1-5 - 06/16/2023: Corn, silage

Field name: K1-5

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.44	2.00	209.04	993.89	180,000.00 *gal*
River	Surface water	0.00	0.00	0.00	66.76	1,920,000.00 *gal*
Application event totals		43.44	2.00	209.04	1,060.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

K1-5 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	63.63	1,830,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.97	3.02	142.70	605.85	120,000.00 *gal*
River	Surface water	0.00	0.00	0.00	62.59	1,800,000.00 *gal*
Application event totals		95.97	3.02	142.70	668.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.97	3.02	142.70	605.85	120,000.00 *gal*
River	Surface water	0.00	0.00	0.00	66.76	1,920,000.00 *gal*
Application event totals		95.97	3.02	142.70	672.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	62.59	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.42	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.42	

K1-5 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	584.15	16,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	584.15	

K1-6 - 12/22/2022: Oats, silage-soft dough

Field name: K1-6

Crop: Oats, silage-soft dough

Plant date: 12/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	154.72	7.27	537.75	2,229.12	720,000.00 *gal*
Application event totals		154.72	7.27	537.75	2,229.12	

04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.75	1.60	167.23	795.11	240,000.00 *gal*
River	Surface water	0.00	0.00	0.00	62.59	3,000,000.00 *gal*
Application event totals		34.75	1.60	167.23	857.70	

K1-6 - 06/16/2023: Corn, silage

Field name: K1-6

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

K1-6 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.75	1.60	167.23	795.11	240,000.00 *gal*
River	Surface water	0.00	0.00	0.00	65.09	3,120,000.00 *gal*
Application event totals		34.75	1.60	167.23	860.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.08	2,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.16	3.63	171.24	727.02	240,000.00 *gal*
River	Surface water	0.00	0.00	0.00	62.59	3,000,000.00 *gal*
Application event totals		115.16	3.63	171.24	789.60	

K1-6 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.77	3.17	149.83	636.14	210,000.00 *gal*
River	Surface water	0.00	0.00	0.00	60.08	2,880,000.00 *gal*
Application event totals		100.77	3.17	149.83	696.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.58	2,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.08	2,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.08	2,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.08	

K1-7 - 12/22/2022: Oats, silage-soft dough

Field name: K1-7

Crop: Oats, silage-soft dough

Plant date: 12/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	154.72	7.27	537.75	2,229.12	1,440,000.00 *gal*
Application event totals		154.72	7.27	537.75	2,229.12	

K1-7 - 12/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.12	2.40	250.85	1,192.67	720,000.00 *gal*
River	Surface water	0.00	0.00	0.00	52.57	5,040,000.00 *gal*
Application event totals		52.12	2.40	250.85	1,245.24	

K1-7 - 06/16/2023: Corn, silage

Field name: K1-7

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.75	1.60	167.23	795.11	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	63.84	6,120,000.00 *gal*
Application event totals		34.75	1.60	167.23	858.95	

06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

K1-7 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.16	3.63	171.24	727.02	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	60.08	5,760,000.00 *gal*
Application event totals		115.16	3.63	171.24	787.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.83	5,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	57.58	5,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.37	2.72	128.43	545.26	360,000.00 *gal*
River	Surface water	0.00	0.00	0.00	56.33	5,400,000.00 *gal*
Application event totals		86.37	2.72	128.43	601.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	55.08	5,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	52.57	5,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.57	

K1-8 - 12/22/2022: Oats, silage-soft dough

Field name: K1-8

Crop: Oats, silage-soft dough

Plant date: 12/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	151.68	7.13	527.21	2,185.41	1,200,000.00 *gal*
Application event totals		151.68	7.13	527.21	2,185.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.10	2.36	245.93	1,169.28	600,000.00 *gal*
River	Surface water	0.00	0.00	0.00	63.32	5,160,000.00 *gal*
Application event totals		51.10	2.36	245.93	1,232.61	

K1-8 - 06/18/2023: Corn, silage

Field name: K1-8

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.88	1.88	196.75	935.43	480,000.00 *gal*
River	Surface water	0.00	0.00	0.00	64.80	5,280,000.00 *gal*
Application event totals		40.88	1.88	196.75	1,000.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	20.00	20.00	0.00	
Application event totals		8.00	20.00	20.00	0.00	

K1-8 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.61	3.20	151.09	641.49	360,000.00 *gal*
River	Surface water	0.00	0.00	0.00	58.91	4,800,000.00 *gal*
Application event totals		101.61	3.20	151.09	700.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	60.38	4,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	58.91	4,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.61	3.20	151.09	641.49	360,000.00 *gal*
River	Surface water	0.00	0.00	0.00	55.96	4,560,000.00 *gal*
Application event totals		101.61	3.20	151.09	697.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	53.02	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.02	

K1-8 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	53.02	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.02	

K1-9 - 11/17/2022: Alfalfa, hay

Field name: K1-9

Crop: Alfalfa, hay

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	265.38	143.51	267.50	5,498.57	550.00 *ton*
Application event totals		265.38	143.51	267.50	5,498.57	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	73.85	7,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.85	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	72.60	6,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.60	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	71.35	6,840,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.35	

K1-9 - 11/17/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	70.10	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.10	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	70.10	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.10	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	6.76	648,000.00 *gal*
Application event totals		0.00	0.00	0.00	6.76	

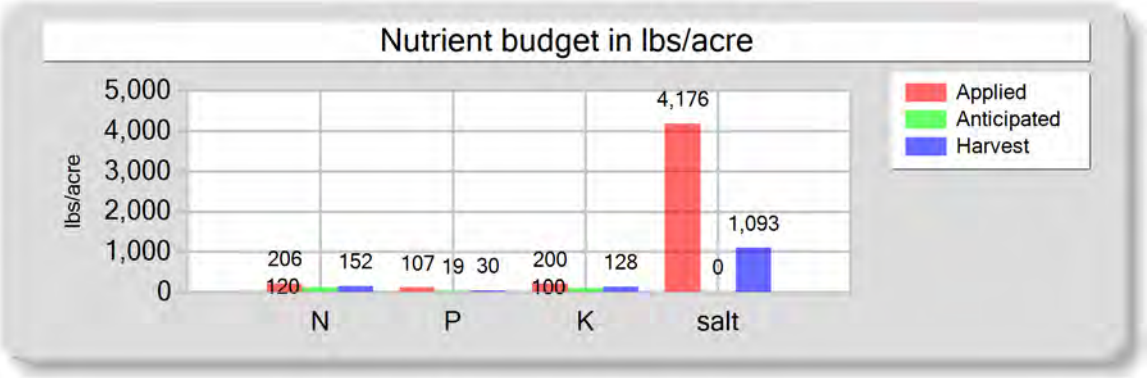
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	67.59	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.59	

B. NUTRIENT BUDGET

C-1 - 12/23/2022: Oats, silage-soft dough

Field name: C-1 Crop: Oats, silage-soft dough Plant date: 12/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	198.59	107.40	200.18	4,114.87
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	60.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.59	107.40	200.18	4,175.83
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	151.55	29.72	128.05	1,092.90
Nutrient balance	54.04	77.68	72.13	3,082.93
Applied to removed ratio	1.36	3.61	1.56	3.82

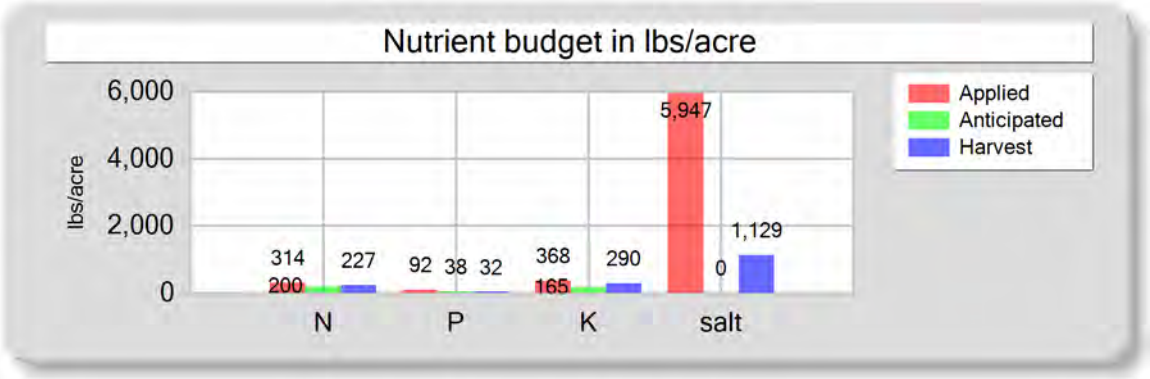
Fresh water applied	
10,080,000.00	*gallons*
371.21	*acre-inches*
5.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C-1 - 06/17/2023: Corn, silage

Field name: C-1 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	20.00	20.00	0.00
Dry manure	234.90	72.38	348.41	5,495.93
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	451.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.90	92.38	368.41	5,947.29
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	227.41	31.52	289.89	1,129.40
Nutrient balance	86.49	60.86	78.52	4,817.89
Applied to removed ratio	1.38	2.93	1.27	5.27

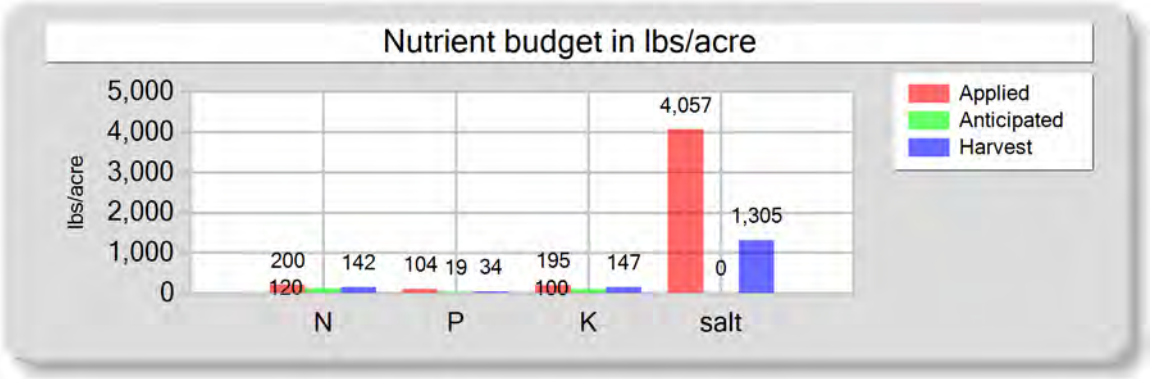
Fresh water applied	
74,640,000.00	*gallons*
2,748.74	*acre-inches*
39.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C-2 - 12/23/2022: Oats, silage-soft dough

Field name: C-2 Crop: Oats, silage-soft dough Plant date: 12/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	193.00	104.37	194.54	3,998.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	58.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.00	104.37	194.54	4,057.15
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	142.37	33.95	147.43	1,305.20
Nutrient balance	57.63	70.42	47.11	2,751.95
Applied to removed ratio	1.40	3.07	1.32	3.11

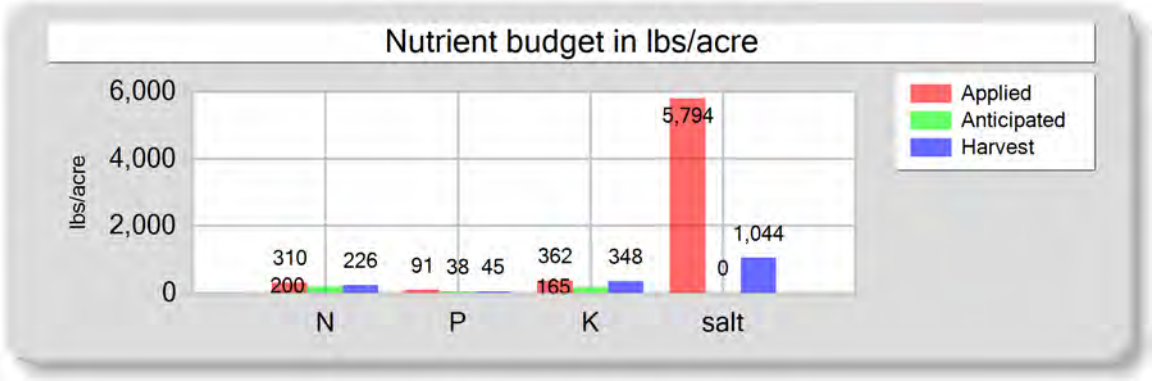
Fresh water applied	
10,320,000.00	*gallons*
380.05	*acre-inches*
5.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C-2 - 06/17/2023: Corn, silage

Field name: C-2 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	20.00	20.00	0.00
Dry manure	230.71	71.09	342.20	5,397.90
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	395.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	309.71	91.09	362.20	5,793.72
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	226.21	44.82	348.38	1,043.55
Nutrient balance	83.50	46.27	13.81	4,750.17
Applied to removed ratio	1.37	2.03	1.04	5.55

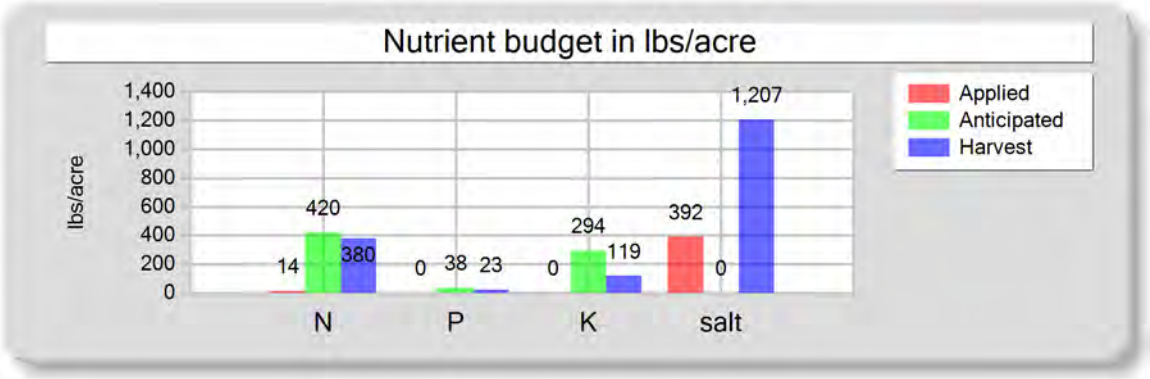
Fresh water applied	
70,200,000.00	*gallons*
2,585.23	*acre-inches*
34.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-1 - 11/17/2021: Alfalfa, hay

Field name: CC-1 Crop: Alfalfa, hay Plant date: 11/17/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	391.52
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	391.52
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	380.32	22.96	119.12	1,206.91
Nutrient balance	-366.32	-22.96	-119.12	-815.39
Applied to removed ratio	0.04	0.00	0.00	0.32

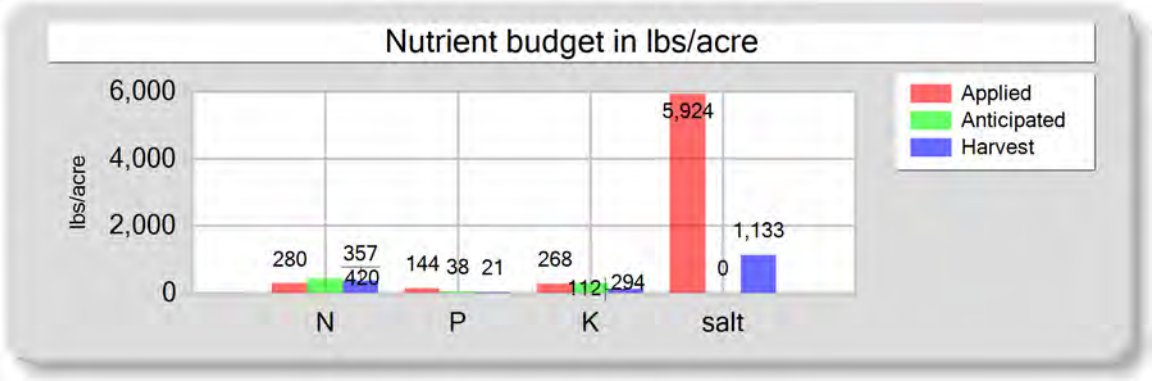
Fresh water applied	
395,040,000.00	*gallons*
14,547.98	*acre-inches*
34.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-2 - 11/09/2022: Alfalfa, hay

Field name: CC-2 Crop: Alfalfa, hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	266.34	144.04	268.47	5,518.56
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	405.57
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	280.34	144.04	268.47	5,924.13
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	357.01	21.02	111.82	1,132.94
Nutrient balance	-76.67	123.02	156.65	4,791.19
Applied to removed ratio	0.79	6.85	2.40	5.23

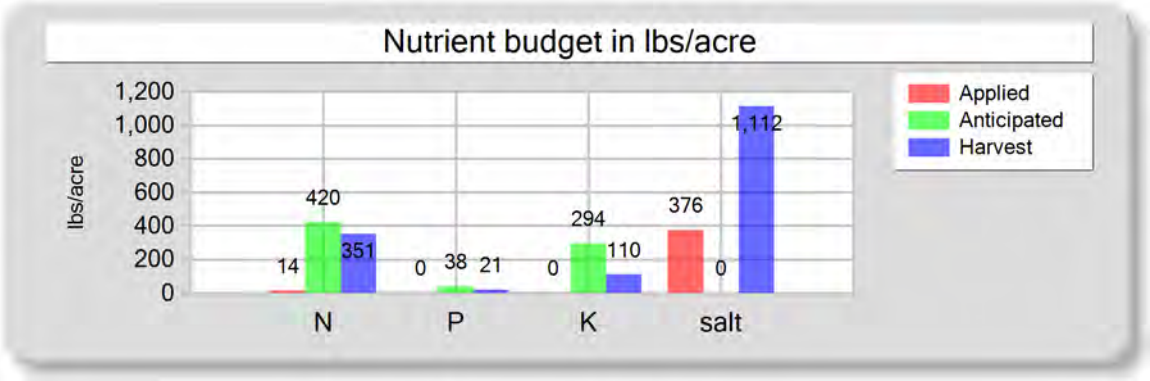
Fresh water applied	
243,000,000.00	*gallons*
8,948.86	*acre-inches*
35.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-3 - 11/12/2017: Alfalfa, hay

Field name: CC-3 Crop: Alfalfa, hay Plant date: 11/12/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	375.87
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	375.87
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	350.54	20.64	109.79	1,112.41
Nutrient balance	-336.54	-20.64	-109.79	-736.54
Applied to removed ratio	0.04	0.00	0.00	0.34

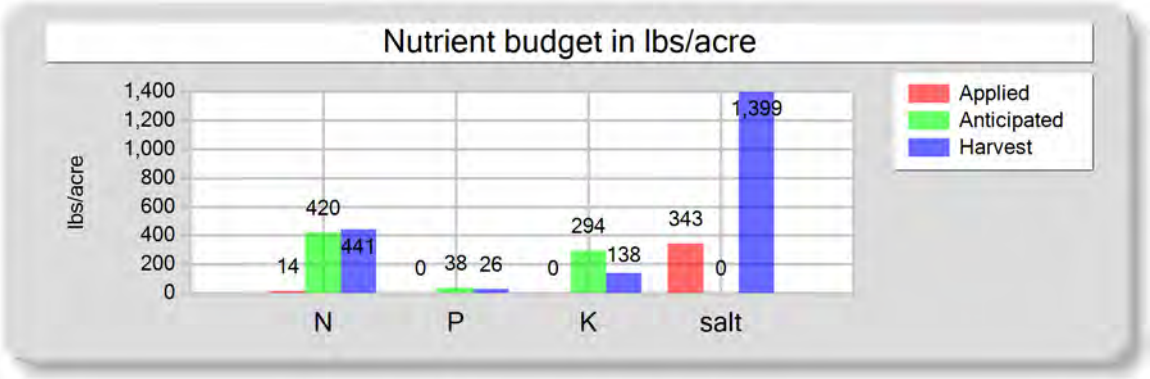
Fresh water applied	
193,680,000.00	*gallons*
7,132.58	*acre-inches*
33.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-4 - 11/06/2018: Alfalfa, hay

Field name: CC-4 Crop: Alfalfa, hay Plant date: 11/06/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	342.77
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	342.77
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	440.83	25.95	138.07	1,398.96
Nutrient balance	-426.83	-25.95	-138.07	-1,056.18
Applied to removed ratio	0.03	0.00	0.00	0.25

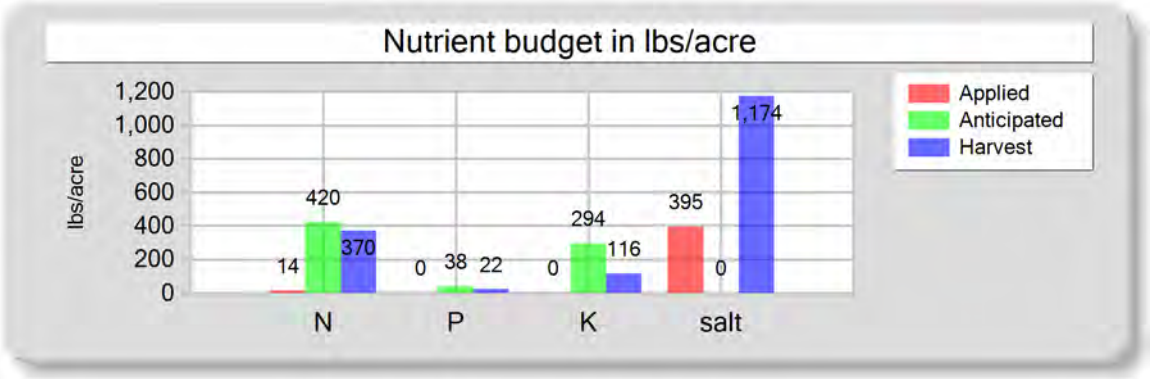
Fresh water applied	
261,240,000.00	*gallons*
9,620.58	*acre-inches*
30.25	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-5 - 11/06/2018: Alfalfa, hay

Field name: CC-5 Crop: Alfalfa, hay Plant date: 11/06/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	395.50
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	395.50
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	370.04	21.78	115.90	1,174.31
Nutrient balance	-356.04	-21.78	-115.90	-778.81
Applied to removed ratio	0.04	0.00	0.00	0.34

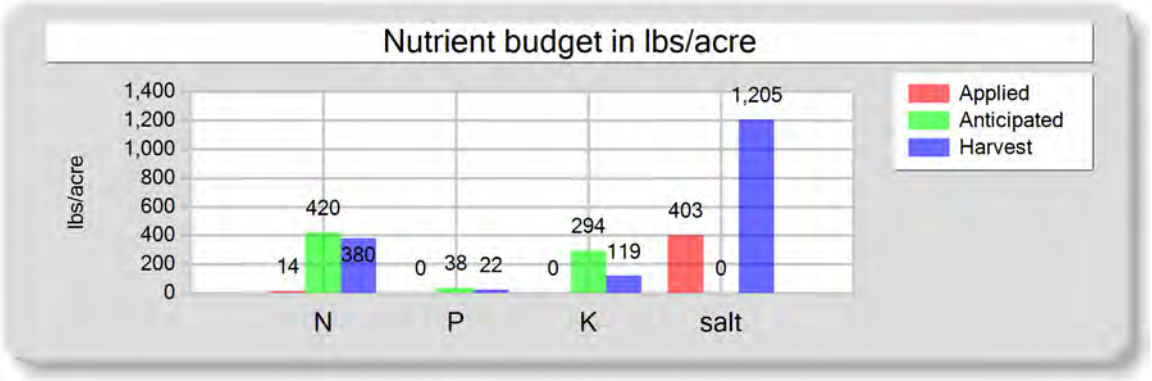
Fresh water applied	
262,560,000.00	*gallons*
9,669.19	*acre-inches*
34.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-6 - 11/22/2019: Alfalfa, hay

Field name: CC-6 Crop: Alfalfa, hay Plant date: 11/22/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	402.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	402.54
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	379.59	22.35	118.89	1,204.60
Nutrient balance	-365.59	-22.35	-118.89	-802.06
Applied to removed ratio	0.04	0.00	0.00	0.33

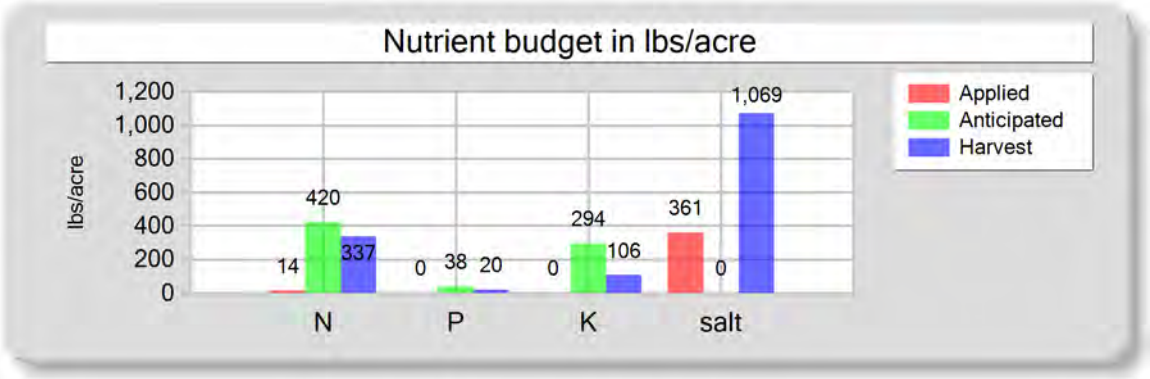
Fresh water applied	
244,080,000.00	*gallons*
8,988.64	*acre-inches*
35.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-7 - 11/22/2019: Alfalfa, hay

Field name: CC-7 Crop: Alfalfa, hay Plant date: 11/22/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	361.09
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	361.09
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	336.96	19.84	105.54	1,069.31
Nutrient balance	-322.96	-19.84	-105.54	-708.22
Applied to removed ratio	0.04	0.00	0.00	0.34

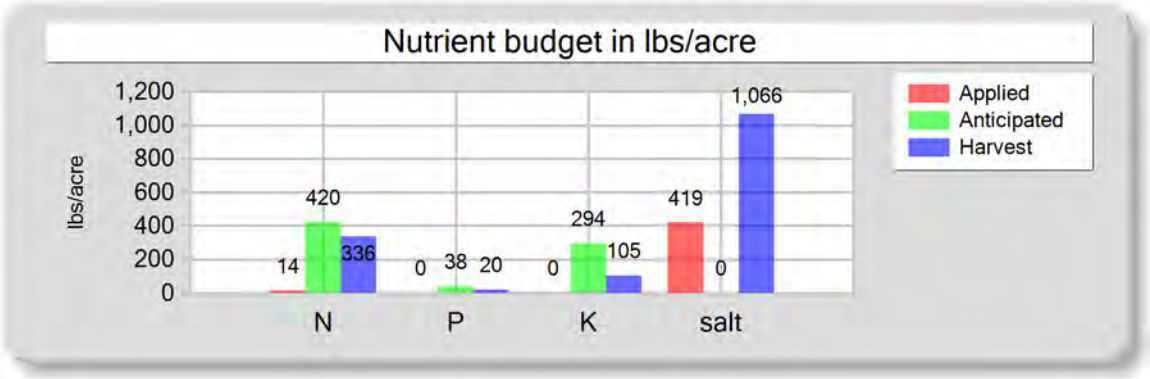
Fresh water applied	
237,120,000.00	*gallons*
8,732.32	*acre-inches*
31.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-8 - 11/23/2019: Alfalfa, hay

Field name: CC-8 Crop: Alfalfa, hay Plant date: 11/23/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	418.97
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	418.97
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	335.93	19.78	105.21	1,066.04
Nutrient balance	-321.93	-19.78	-105.21	-647.07
Applied to removed ratio	0.04	0.00	0.00	0.39

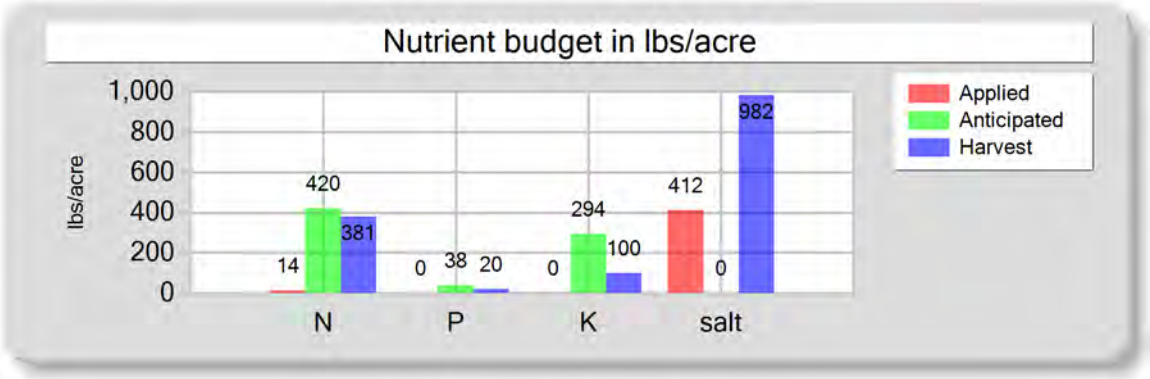
Fresh water applied	
136,560,000.00	*gallons*
5,029.04	*acre-inches*
36.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-B2 - 10/24/2020: Alfalfa, hay

Field name: CC-B2 Crop: Alfalfa, hay Plant date: 10/24/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	412.30
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	412.30
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	380.70	19.83	100.46	982.21
Nutrient balance	-366.70	-19.83	-100.46	-569.91
Applied to removed ratio	0.04	0.00	0.00	0.42

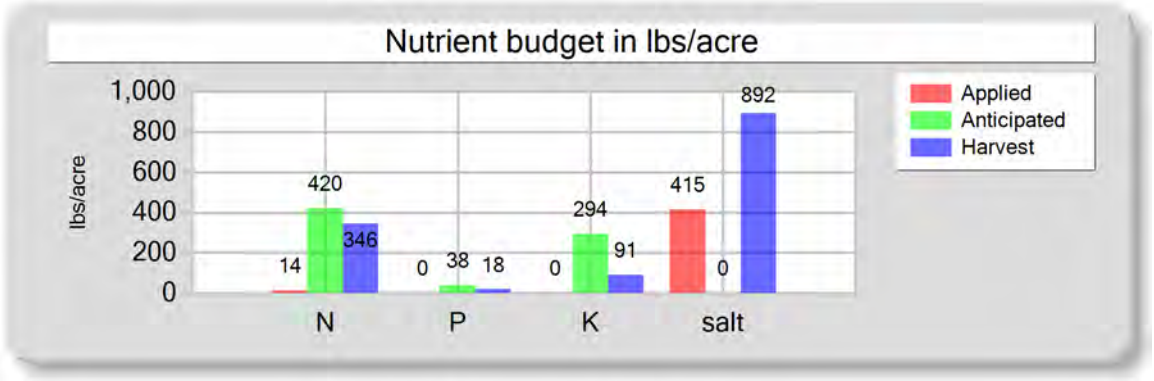
Fresh water applied	
63,240,000.00	*gallons*
2,328.91	*acre-inches*
36.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-B3 - 10/24/2020: Alfalfa, hay

Field name: CC-B3 Crop: Alfalfa, hay Plant date: 10/24/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	415.29
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	415.29
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	345.60	18.24	90.96	891.65
Nutrient balance	-331.60	-18.24	-90.96	-476.36
Applied to removed ratio	0.04	0.00	0.00	0.47

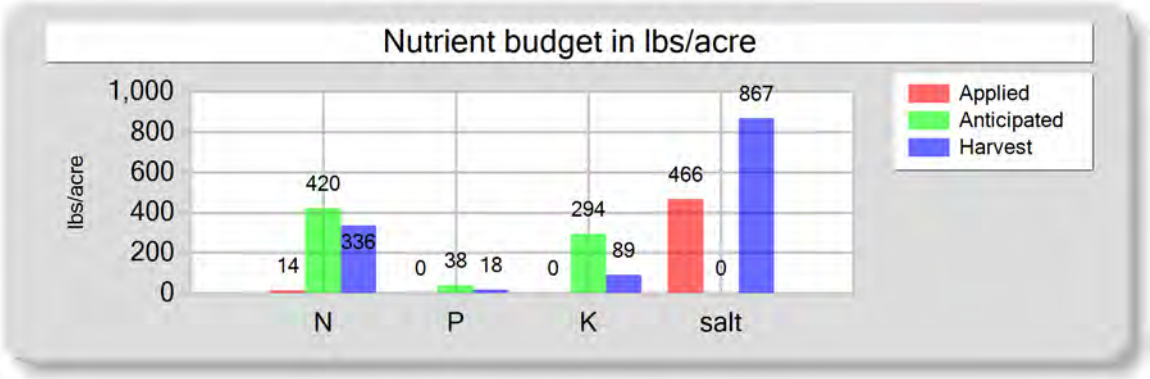
Fresh water applied	
50,760,000.00	*gallons*
1,869.32	*acre-inches*
36.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-D5 - 10/25/2020: Alfalfa, hay

Field name: CC-D5 Crop: Alfalfa, hay Plant date: 10/25/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	466.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	466.13
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	336.00	17.50	88.67	866.88
Nutrient balance	-322.00	-17.50	-88.67	-400.75
Applied to removed ratio	0.04	0.00	0.00	0.54

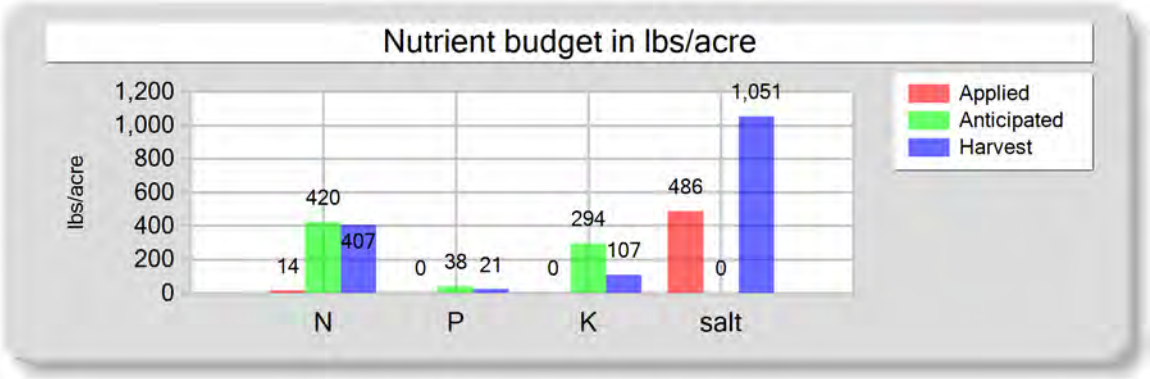
Fresh water applied	
46,920,000.00	*gallons*
1,727.90	*acre-inches*
41.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-P8 - 10/25/2020: Alfalfa, hay

Field name: CC-P8 Crop: Alfalfa, hay Plant date: 10/25/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	486.39
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	486.39
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	407.31	21.50	107.20	1,050.87
Nutrient balance	-393.31	-21.50	-107.20	-564.48
Applied to removed ratio	0.03	0.00	0.00	0.46

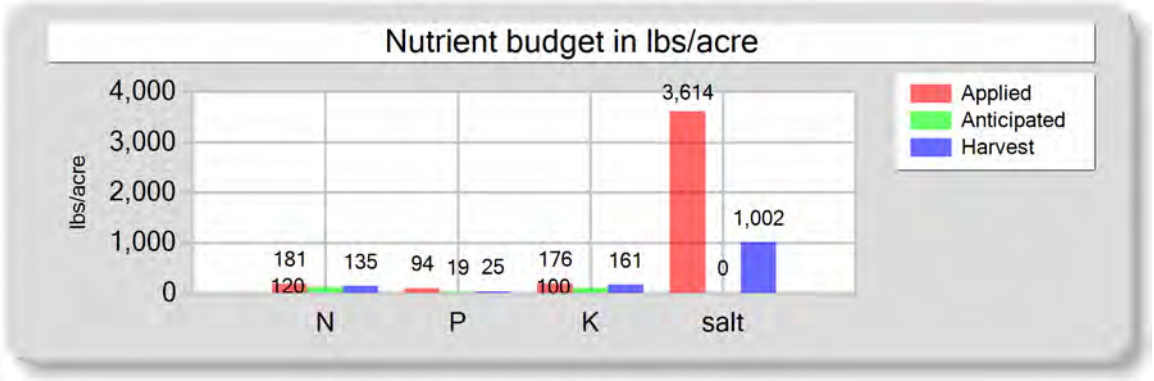
Fresh water applied	
32,640,000.00	*gallons*
1,202.02	*acre-inches*
42.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-W1 - 12/27/2022: Oats, silage-soft dough

Field name: CC-W1 Crop: Oats, silage-soft dough Plant date: 12/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	174.44	94.34	175.84	3,614.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.44	94.34	175.84	3,614.44
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	135.19	25.42	160.61	1,002.41
Nutrient balance	46.25	68.92	15.23	2,612.03
Applied to removed ratio	1.34	3.71	1.09	3.61

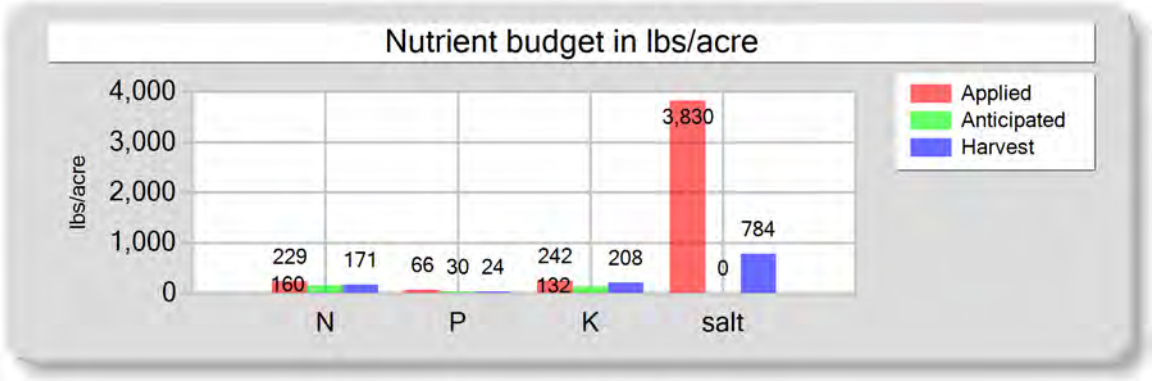
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

CC-W1 - 06/18/2023: Corn, silage

Field name: CC-W1 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	20.00	20.00	0.00
Dry manure	149.61	46.10	221.91	3,500.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	329.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.61	66.10	241.91	3,829.79
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	171.03	24.50	207.55	783.88
Nutrient balance	57.58	41.60	34.36	3,045.92
Applied to removed ratio	1.34	2.70	1.17	4.89

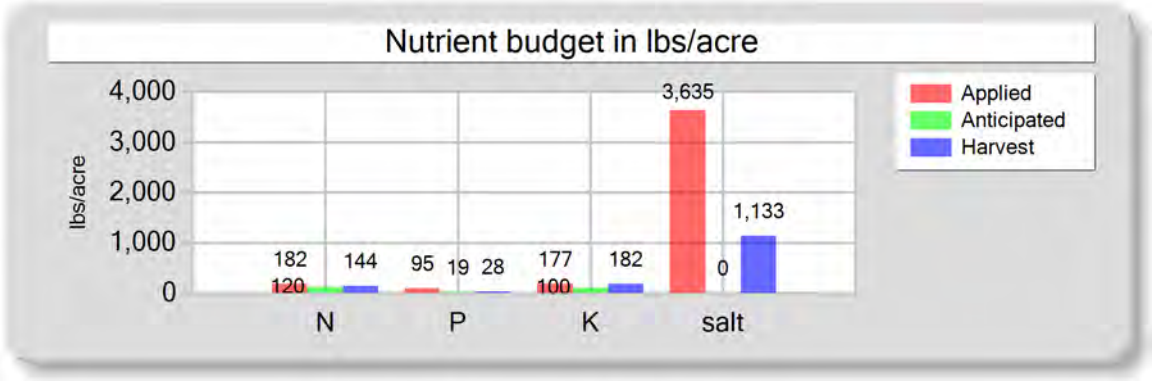
Fresh water applied	
41,040,000.00	*gallons*
1,511.36	*acre-inches*
29.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

CC-W2 - 12/27/2022: Oats, silage-soft dough

Field name: CC-W2 Crop: Oats, silage-soft dough Plant date: 12/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	175.45	94.89	176.86	3,635.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.45	94.89	176.86	3,635.42
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	144.14	28.09	181.94	1,133.25
Nutrient balance	38.32	66.79	-5.09	2,502.17
Applied to removed ratio	1.27	3.38	0.97	3.21

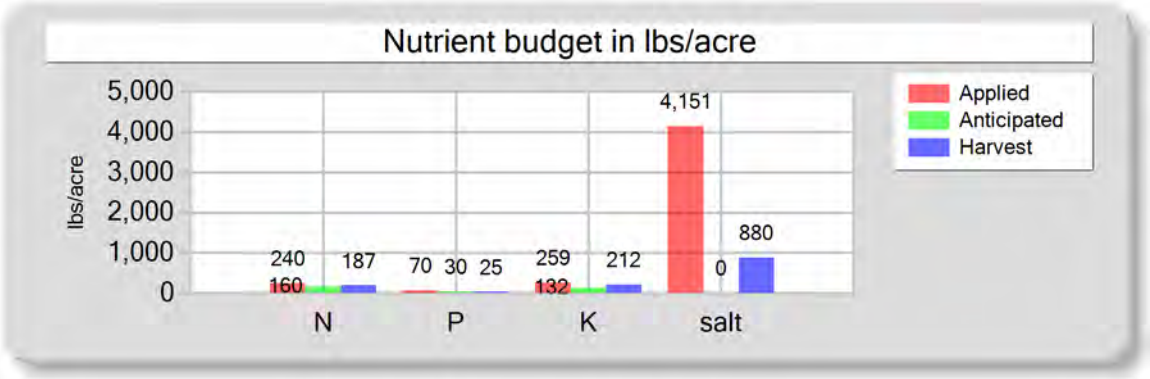
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

CC-W2 - 06/18/2023: Corn, silage

Field name: CC-W2 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	20.00	20.00	0.00
Dry manure	161.10	49.64	238.95	3,769.21
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	381.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.10	69.64	258.95	4,150.65
Anticipated crop nutrient removal	160.00	30.00	132.00	0.00
Actual crop nutrient removal	186.92	25.26	212.18	879.60
Nutrient balance	53.17	44.38	46.76	3,271.06
Applied to removed ratio	1.28	2.76	1.22	4.72

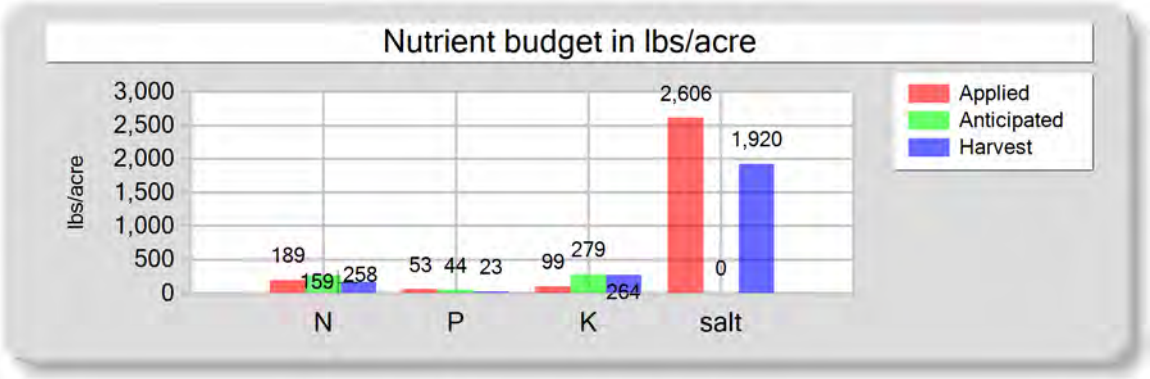
Fresh water applied	
50,280,000.00	*gallons*
1,851.64	*acre-inches*
33.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Escalon - 02/01/2012: Almond, in shell

Field name: Escalon Crop: Almond, in shell Plant date: 02/01/2012



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	76.80	0.00	0.00	0.00
Dry manure	98.64	53.35	99.43	2,043.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	562.14
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	189.44	53.35	99.43	2,606.06
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	158.91	23.43	263.95	1,920.40
Nutrient balance	30.54	29.91	-164.51	685.66
Applied to removed ratio	1.19	2.28	0.38	1.36

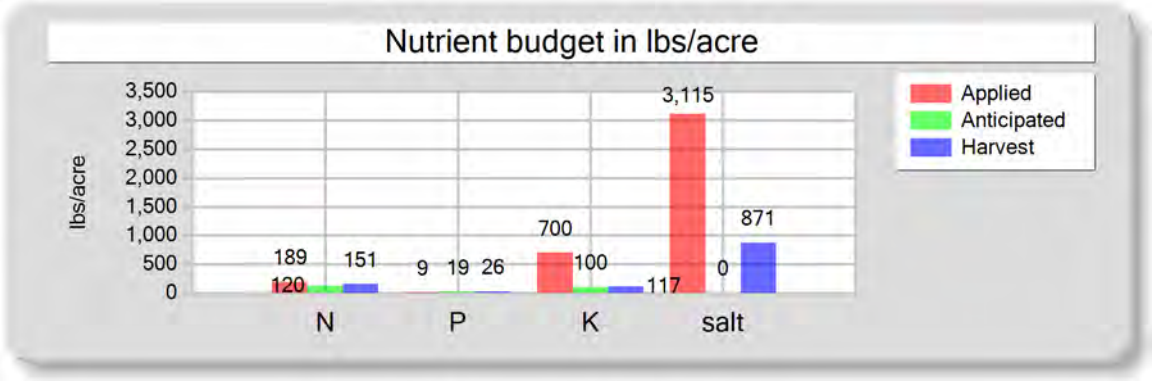
Fresh water applied	
60,626,590.00	*gallons*
2,232.67	*acre-inches*
49.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-1 - 12/21/2022: Oats, silage-soft dough

Field name: K1-1 Crop: Oats, silage-soft dough Plant date: 12/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.25	8.52	699.84	3,046.54
Fresh water	0.00	0.00	0.00	68.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.25	8.52	699.84	3,115.21
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	151.39	25.57	116.99	871.47
Nutrient balance	37.87	-17.05	582.85	2,243.75
Applied to removed ratio	1.25	0.33	5.98	3.57

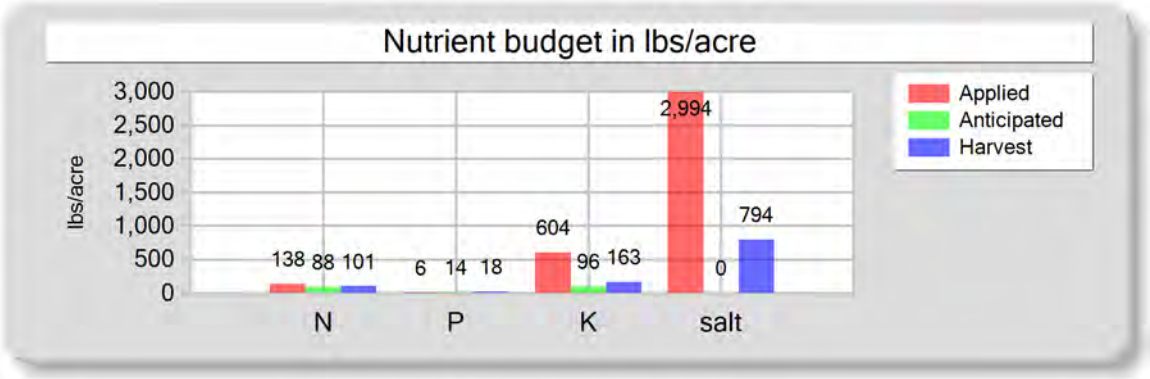
Fresh water applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
6.06	*inches/acre*

Process wastewater applied	
1,680,000.00	*gallons*
61.87	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-1 - 09/02/2023: Sudangrass, silage

Field name: K1-1 Crop: Sudangrass, silage Plant date: 09/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	130.70	5.88	604.27	2,792.48
Fresh water	0.00	0.00	0.00	201.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.70	5.88	604.27	2,994.19
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	101.27	18.36	163.13	793.73
Nutrient balance	36.43	-12.48	441.14	2,200.46
Applied to removed ratio	1.36	0.32	3.70	3.77

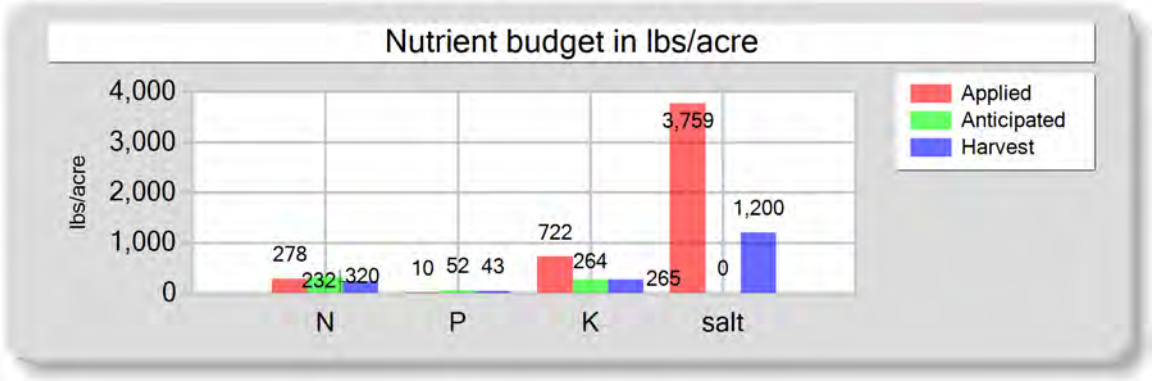
Fresh water applied	
16,920,000.00	*gallons*
623.11	*acre-inches*
17.80	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-10 - 03/13/2021: Oats, hay

Field name: K1-10 Crop: Oats, hay Plant date: 03/13/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.80	9.76	722.44	3,310.34
Fresh water	0.00	0.00	0.00	448.84
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	277.80	9.76	722.44	3,759.18
Anticipated crop nutrient removal	320.00	52.00	264.00	0.00
Actual crop nutrient removal	232.08	43.14	264.83	1,199.99
Nutrient balance	45.72	-33.38	457.61	2,559.20
Applied to removed ratio	1.20	0.23	2.73	3.13

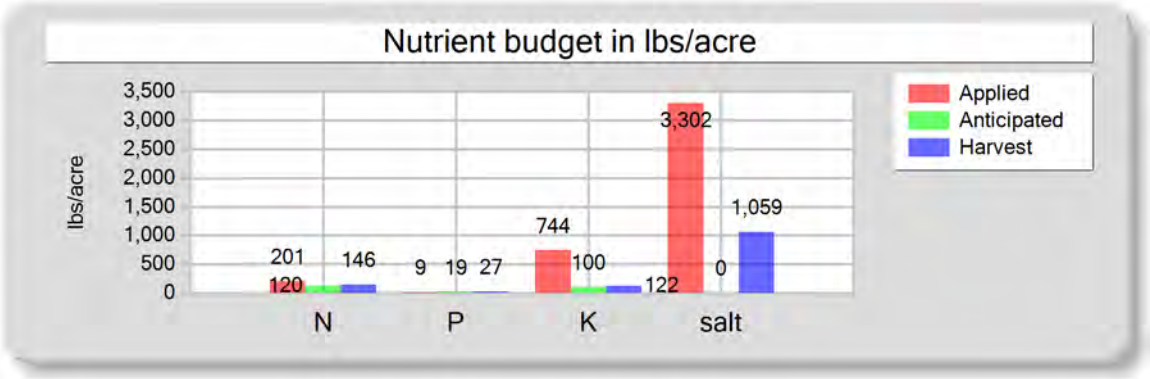
Fresh water applied	
90,360,000.00	*gallons*
3,327.65	*acre-inches*
39.61	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
1.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-2 - 12/21/2022: Oats, silage-soft dough

Field name: K1-2 Crop: Oats, silage-soft dough Plant date: 12/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.95	9.07	743.79	3,236.02
Fresh water	0.00	0.00	0.00	66.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.95	9.07	743.79	3,302.37
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	146.23	27.16	121.74	1,059.47
Nutrient balance	54.72	-18.09	622.05	2,242.89
Applied to removed ratio	1.37	0.33	6.11	3.12

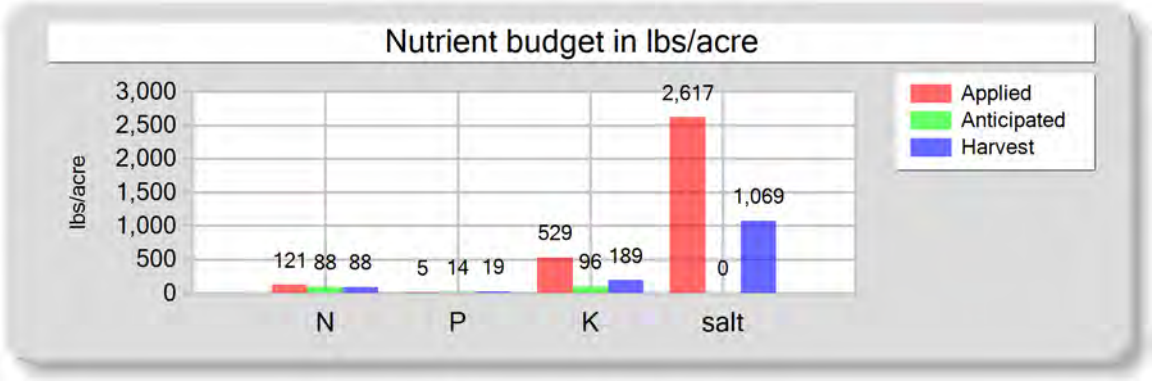
Fresh water applied	
6,360,000.00	*gallons*
234.22	*acre-inches*
5.86	*inches/acre*

Process wastewater applied	
2,040,000.00	*gallons*
75.13	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-2 - 09/02/2023: Sudangrass, silage

Field name: K1-2 Crop: Sudangrass, silage Plant date: 09/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	114.36	5.15	528.74	2,443.42
Fresh water	0.00	0.00	0.00	173.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	121.36	5.15	528.74	2,617.41
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	88.12	19.33	188.55	1,069.35
Nutrient balance	33.24	-14.18	340.19	1,548.06
Applied to removed ratio	1.38	0.27	2.80	2.45

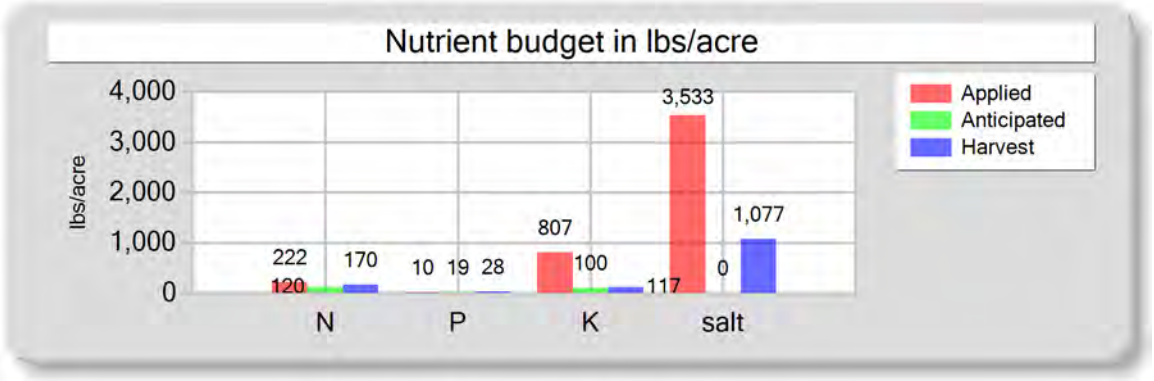
Fresh water applied	
16,680,000.00	*gallons*
614.27	*acre-inches*
15.36	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-3 - 12/21/2022: Oats, silage-soft dough

Field name: K1-3 Crop: Oats, silage-soft dough Plant date: 12/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.34	10.08	806.54	3,470.69
Fresh water	0.00	0.00	0.00	62.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.34	10.08	806.54	3,533.27
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	170.08	28.00	116.97	1,076.99
Nutrient balance	52.26	-17.92	689.58	2,456.28
Applied to removed ratio	1.31	0.36	6.90	3.28

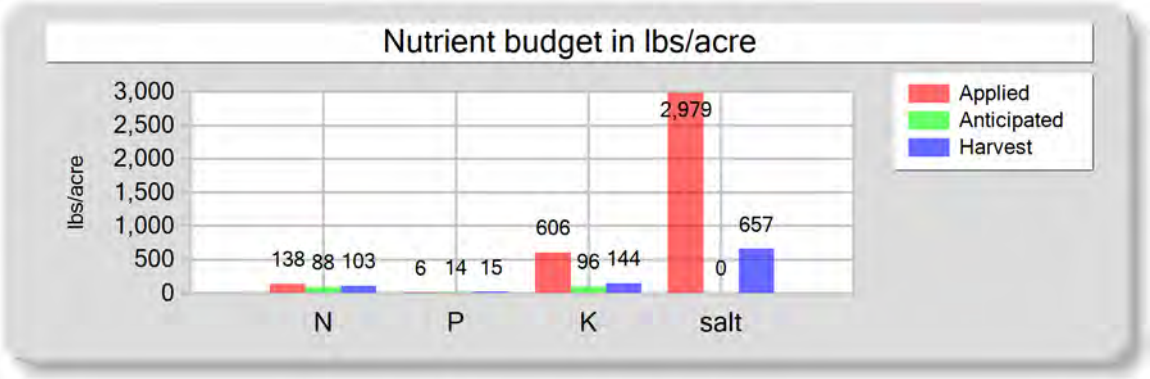
Fresh water applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
5.52	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-3 - 09/02/2023: Sudangrass, silage

Field name: K1-3 Crop: Sudangrass, silage Plant date: 09/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	131.04	5.90	605.85	2,799.75
Fresh water	0.00	0.00	0.00	179.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.04	5.90	605.85	2,979.17
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	103.06	15.49	143.87	656.75
Nutrient balance	34.98	-9.60	461.98	2,322.41
Applied to removed ratio	1.34	0.38	4.21	4.54

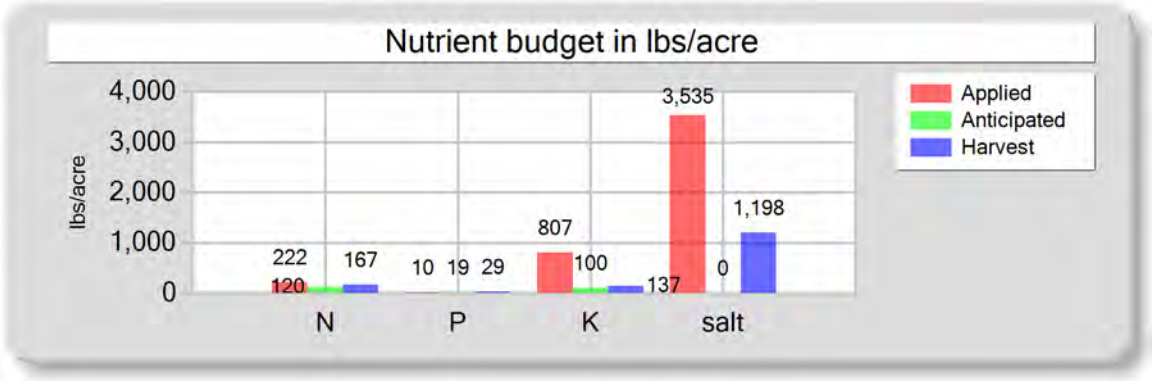
Fresh water applied	
5,160,000.00	*gallons*
190.03	*acre-inches*
15.84	*inches/acre*

Process wastewater applied	
330,000.00	*gallons*
12.15	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-5 - 12/22/2022: Oats, silage-soft dough

Field name: K1-5 Crop: Oats, silage-soft dough Plant date: 12/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.34	10.08	806.54	3,470.69
Fresh water	0.00	0.00	0.00	64.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.34	10.08	806.54	3,535.36
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	166.60	28.56	136.92	1,198.40
Nutrient balance	55.74	-18.48	669.62	2,336.96
Applied to removed ratio	1.33	0.35	5.89	2.95

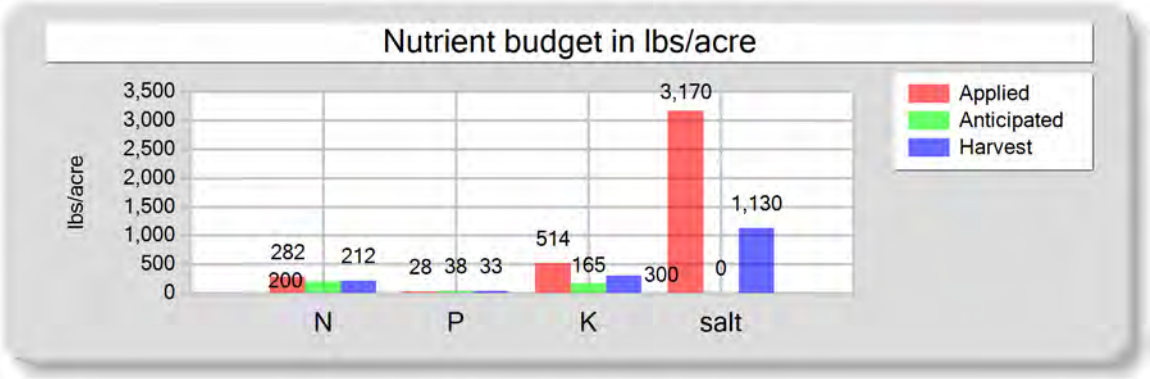
Fresh water applied	
1,860,000.00	*gallons*
68.50	*acre-inches*
5.71	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-5 - 06/16/2023: Corn, silage

Field name: K1-5 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	20.00	20.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.37	8.04	494.44	2,205.58
Fresh water	0.00	0.00	0.00	964.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.37	28.04	514.44	3,170.47
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	211.64	32.76	299.52	1,129.75
Nutrient balance	70.73	-4.72	214.92	2,040.72
Applied to removed ratio	1.33	0.86	1.72	2.81

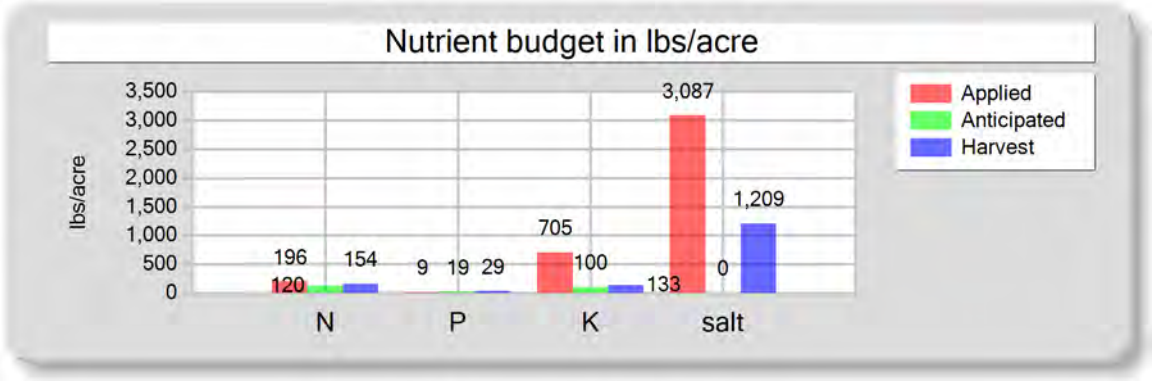
Fresh water applied	
27,750,000.00	*gallons*
1,021.94	*acre-inches*
85.16	*inches/acre*

Process wastewater applied	
420,000.00	*gallons*
15.47	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-6 - 12/22/2022: Oats, silage-soft dough

Field name: K1-6 Crop: Oats, silage-soft dough Plant date: 12/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.46	8.87	704.99	3,024.23
Fresh water	0.00	0.00	0.00	62.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.46	8.87	704.99	3,086.82
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	153.79	28.51	133.34	1,208.91
Nutrient balance	42.67	-19.64	571.64	1,877.91
Applied to removed ratio	1.28	0.31	5.29	2.55

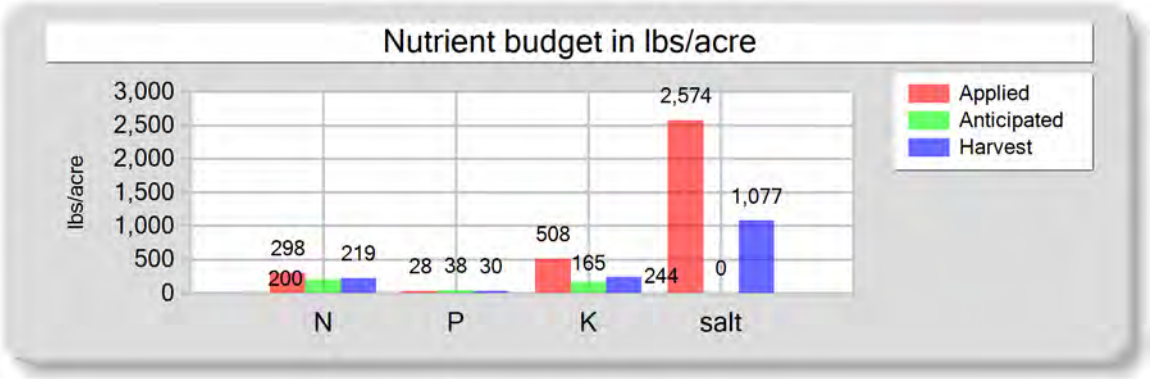
Fresh water applied	
3,000,000.00	*gallons*
110.48	*acre-inches*
5.52	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-6 - 06/16/2023: Corn, silage

Field name: K1-6 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	20.00	20.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	250.68	8.40	488.31	2,158.27
Fresh water	0.00	0.00	0.00	415.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.68	28.40	508.31	2,573.85
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	218.62	29.70	243.71	1,077.15
Nutrient balance	79.05	-1.30	264.60	1,496.70
Applied to removed ratio	1.36	0.96	2.09	2.39

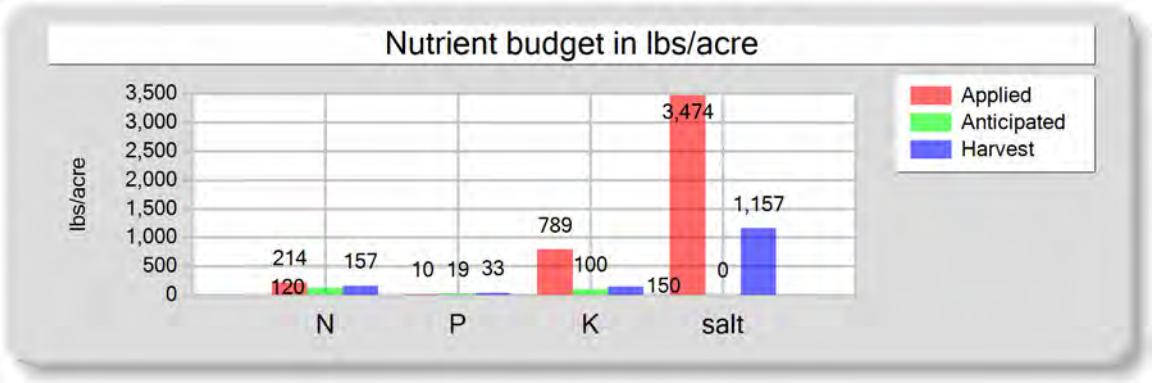
Fresh water applied	
19,920,000.00	*gallons*
733.59	*acre-inches*
36.68	*inches/acre*

Process wastewater applied	
690,000.00	*gallons*
25.41	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-7 - 12/22/2022: Oats, silage-soft dough

Field name: K1-7 Crop: Oats, silage-soft dough Plant date: 12/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.84	9.67	788.60	3,421.78
Fresh water	0.00	0.00	0.00	52.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.84	9.67	788.60	3,474.36
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	156.56	33.36	150.25	1,156.59
Nutrient balance	57.28	-23.68	638.35	2,317.77
Applied to removed ratio	1.37	0.29	5.25	3.00

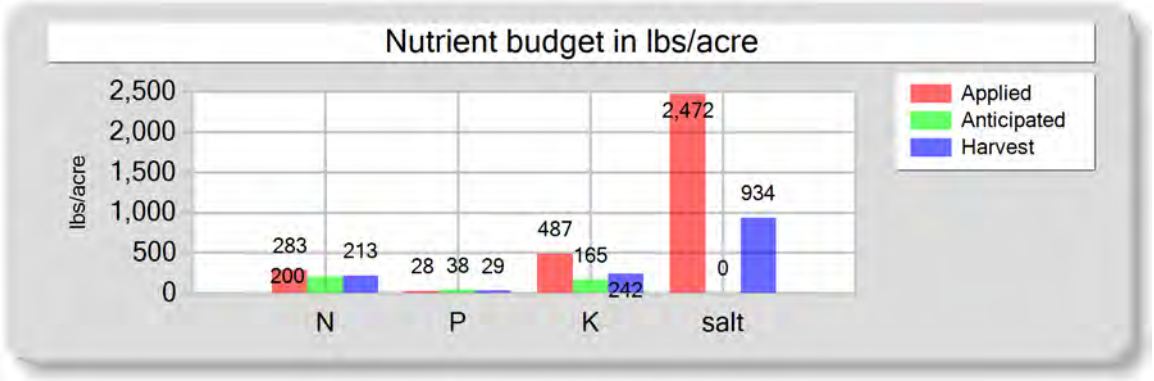
Fresh water applied	
5,040,000.00	*gallons*
185.61	*acre-inches*
4.64	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-7 - 06/16/2023: Corn, silage

Field name: K1-7 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	20.00	20.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.28	7.95	466.90	2,067.39
Fresh water	0.00	0.00	0.00	404.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.28	27.95	486.90	2,471.71
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	213.47	28.85	242.32	934.03
Nutrient balance	69.81	-0.90	244.58	1,537.68
Applied to removed ratio	1.33	0.97	2.01	2.65

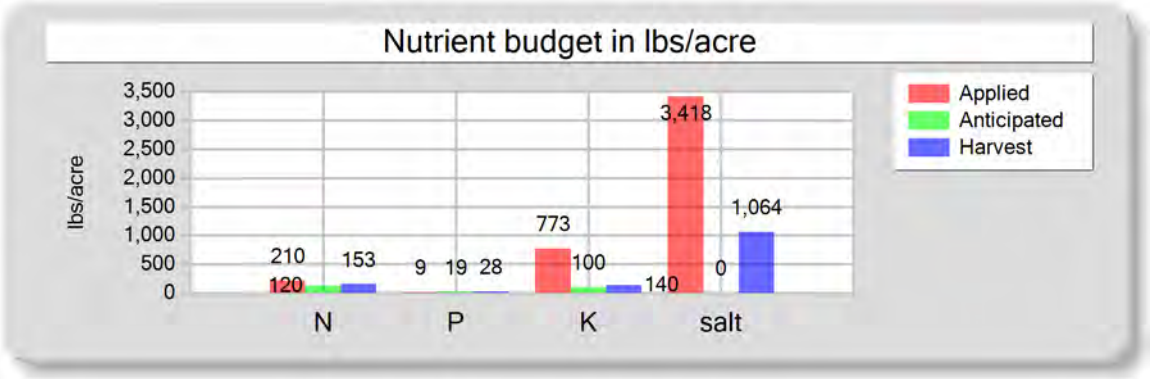
Fresh water applied	
38,760,000.00	*gallons*
1,427.40	*acre-inches*
35.68	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-8 - 12/22/2022: Oats, silage-soft dough

Field name: K1-8 Crop: Oats, silage-soft dough Plant date: 12/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.78	9.48	773.14	3,354.69
Fresh water	0.00	0.00	0.00	63.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.78	9.48	773.14	3,418.01
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	153.23	28.14	139.52	1,063.69
Nutrient balance	56.55	-18.66	633.62	2,354.32
Applied to removed ratio	1.37	0.34	5.54	3.21

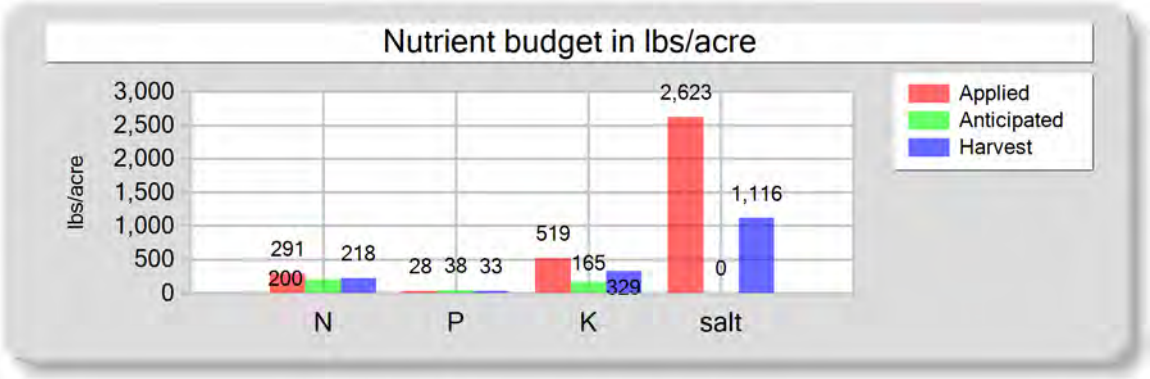
Fresh water applied	
5,160,000.00	*gallons*
190.03	*acre-inches*
5.59	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-8 - 06/18/2023: Corn, silage

Field name: K1-8 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	20.00	20.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	244.11	8.28	498.93	2,218.40
Fresh water	0.00	0.00	0.00	404.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.11	28.28	518.93	2,623.37
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	218.03	33.04	328.70	1,115.71
Nutrient balance	73.07	-4.75	190.23	1,507.66
Applied to removed ratio	1.34	0.86	1.58	2.35

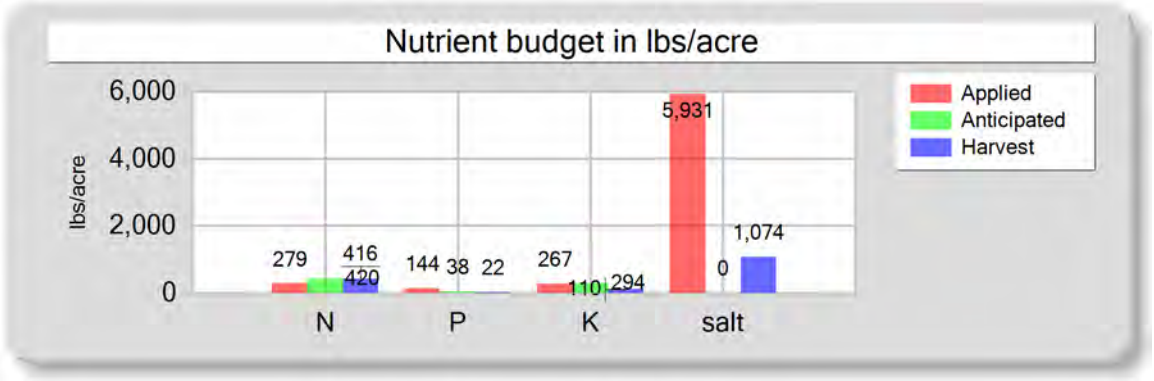
Fresh water applied	
33,000,000.00	*gallons*
1,215.28	*acre-inches*
35.74	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
1.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

K1-9 - 11/17/2022: Alfalfa, hay

Field name: K1-9 Crop: Alfalfa, hay Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	265.38	143.51	267.50	5,498.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	432.35
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	279.38	143.51	267.50	5,930.92
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	416.16	21.96	109.53	1,073.69
Nutrient balance	-136.79	121.55	157.97	4,857.23
Applied to removed ratio	0.67	6.53	2.44	5.52

Fresh water applied	
41,448,000.00	*gallons*
1,526.39	*acre-inches*
38.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	6,760.00	12,600.00							25.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	6,440.00	31,000.00							48.90
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/28/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	3,480.00	8,190.00							11.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	7,750.00	40,800.00							51.70
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	5,050.00	21,300.00							27.30
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 01/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	515.00	246.00	0.00	0.00	24.20	1,790.00								12,300.00	7,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	347.00	171.00	0.00	0.00	16.00	1,670.00	106.00	269.00	910.00	3,280.00	0.00	294.00	72.40	11,900.00	7,940
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,150.00	706.00	0.00	0.00	36.20	1,710.00								14,700.00	7,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	571.00	263.00	0.00	0.00	25.70	2,640.00								18,200.00	12,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal SSJID

Canal SSJID

Canal SSJID

Sample description: Canal SSJID

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								89.20	58
DL	1.00	0.50	0.50								10.00	10

River

River

Sample description: River

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								168.00	50
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

C-1 - 12/23/2022: Oats, silage-soft dough

C-1 - 12/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,610.00	1,100.00	4,740.00		10.40
DL	100.00	100.00	100.00		0.01

C-1 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	560.00	5,150.00		6.60
DL	100.00	100.00	100.00		0.01

C-2 - 12/23/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,780.00	1,140.00	4,950.00		9.76
DL	100.00	100.00	100.00		0.01

C-2 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,240.00	840.00	6,530.00		6.00
DL	100.00	100.00	100.00		0.01

CC-2 - 11/09/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-3 - 11/12/2017: Alfalfa, hay

CC-3 - 11/12/2017: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-4 - 11/06/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-5 - 11/06/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-6 - 11/22/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-7 - 11/22/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-8 - 11/23/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	1,560.00	8,300.00		9.60
DL	100.00	100.00	100.00		0.01

CC-B2 - 10/24/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,500.00	7,600.00		8.60
DL	100.00	100.00	100.00		0.01

CC-B3 - 10/24/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,520.00	7,580.00		8.60
DL	100.00	100.00	100.00		0.01

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,520.00	7,580.00		8.60
DL	100.00	100.00	100.00		0.01

CC-D5 - 10/25/2020: Alfalfa, hay

CC-D5 - 10/25/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,500.00	7,600.00		8.60
DL	100.00	100.00	100.00		0.01

CC-P8 - 10/25/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,520.00	7,580.00		8.60
DL	100.00	100.00	100.00		0.01

CC-W1 - 12/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	940.00	5,940.00		9.86
DL	100.00	100.00	100.00		0.01

CC-W1 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,840.00	550.00	4,660.00		5.50
DL	100.00	100.00	100.00		0.01

CC-W2 - 12/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,490.00	1,070.00	6,930.00		10.90
DL	100.00	100.00	100.00		0.01

CC-W2 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	550.00	4,620.00		5.70
DL	100.00	100.00	100.00		0.01

Escalon - 02/01/2012: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	1,740.00	19,600.00		15.40
DL	100.00	100.00	100.00		0.01

K1-1 - 12/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,690.00	1,130.00	5,170.00		7.99
DL	100.00	100.00	100.00		0.01

K1-1 - 09/02/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 11/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,680.00	1,030.00	9,150.00		14.00
DL	100.00	100.00	100.00		0.01

K1-10 - 03/13/2021: Oats, hay

Orchard Grass

Sample and source description: Orchard Grass

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 14.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,030.00	18,600.00		9.80
DL	100.00	100.00	100.00		0.01

K1-2 - 12/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,030.00	1,120.00	5,020.00		9.84
DL	100.00	100.00	100.00		0.01

K1-2 - 09/02/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 11/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,650.00	1,020.00	9,950.00		17.10
DL	100.00	100.00	100.00		0.01

K1-3 - 12/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,500.00	1,070.00	4,470.00		9.27
DL	100.00	100.00	100.00		0.01

K1-3 - 09/02/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 11/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,450.00	1,120.00	10,400.00		14.30
DL	100.00	100.00	100.00		0.01

K1-5 - 12/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,950.00	1,020.00	4,890.00		10.70
DL	100.00	100.00	100.00		0.01

K1-5 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	630.00	5,760.00		7.10
DL	100.00	100.00	100.00		0.01

K1-6 - 12/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,340.00	990.00	4,630.00		10.60
DL	100.00	100.00	100.00		0.01

K1-6 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,270.00	580.00	4,760.00		6.70
DL	100.00	100.00	100.00		0.01

K1-7 - 12/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,210.00	1,110.00	5,000.00		9.48
DL	100.00	100.00	100.00		0.01

K1-7 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	550.00	4,620.00		5.30
DL	100.00	100.00	100.00		0.01

K1-8 - 12/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,480.00	1,190.00	5,900.00		10.20
DL	100.00	100.00	100.00		0.01

K1-8 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	600.00	5,970.00		6.80
DL	100.00	100.00	100.00		0.01

K1-9 - 11/17/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 13.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,800.00	1,520.00	7,580.00		8.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

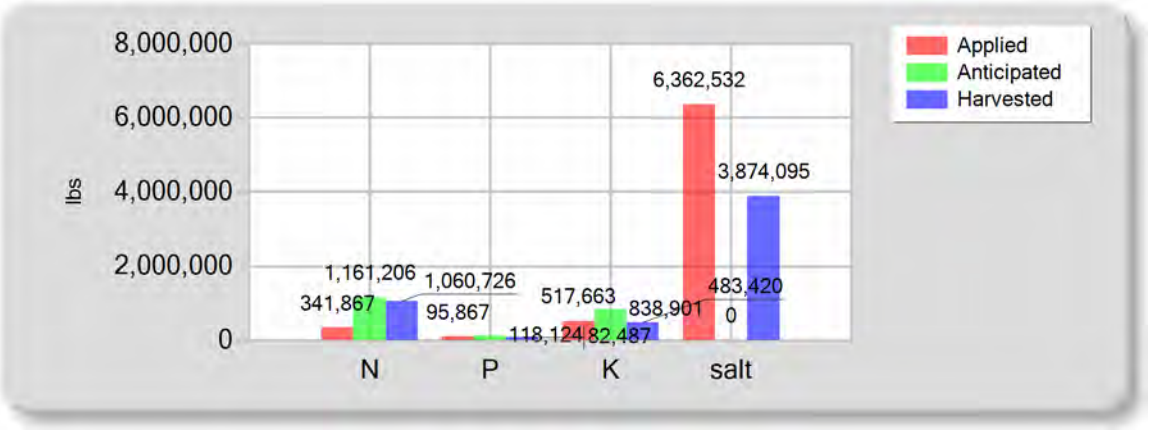
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

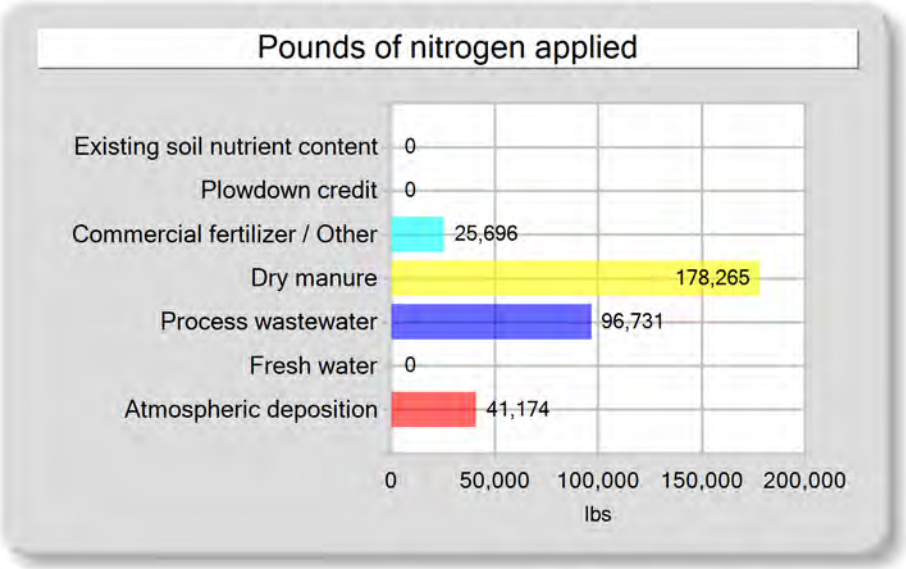
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

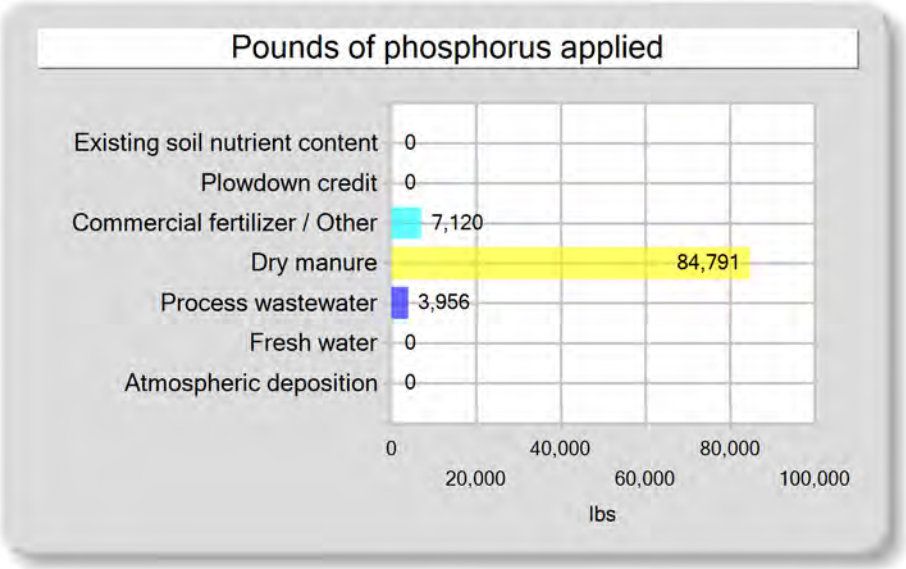
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25,696.00	7,120.00	7,120.00	0.00
Dry manure	178,265.49	84,791.17	203,416.50	3,827,304.46
Process wastewater	96,731.48	3,955.76	307,126.88	1,365,674.27
Fresh water	0.00	0.00	0.00	1,169,553.67
Atmospheric deposition	41,174.00	0.00	0.00	0.00
Total nutrients applied	341,866.97	95,866.93	517,663.38	6,362,532.40
Anticipated crop nutrient removal	1,161,206.00	118,123.50	838,900.80	0.00
Actual crop nutrient removal	1,060,725.88	82,486.72	483,419.64	3,874,095.40
Nutrient balance	-718,858.91	13,380.21	34,243.74	2,488,436.99
Applied to removed ratio	0.32	1.16	1.07	1.64

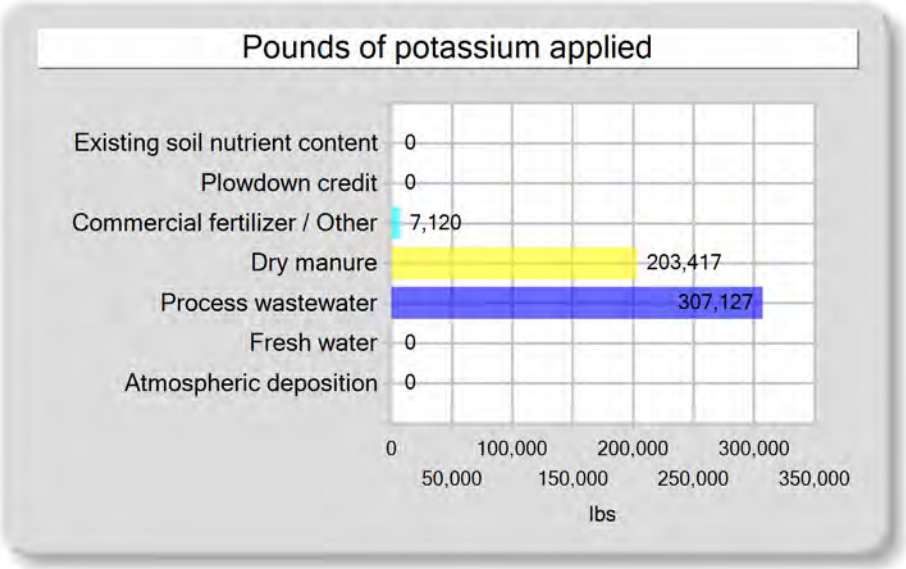
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

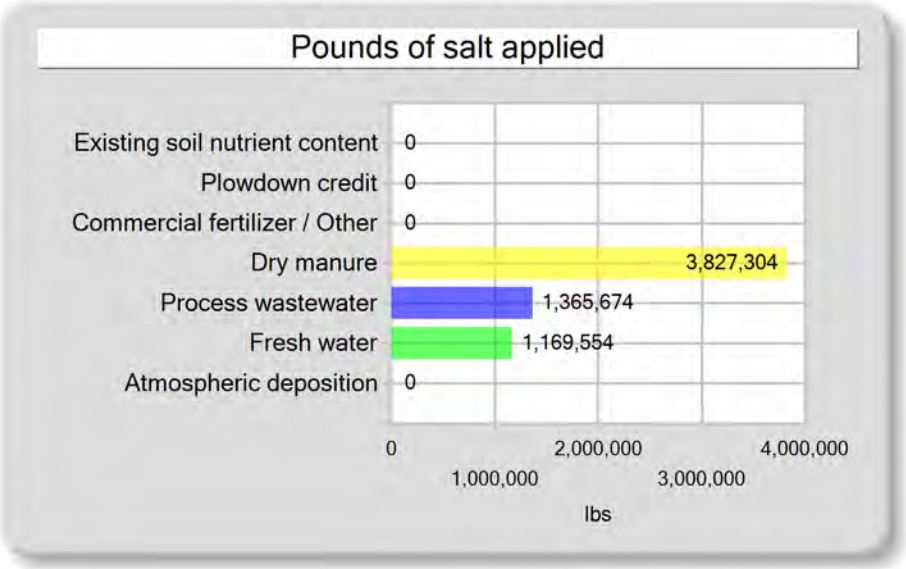


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

Discharge date	Location	Map reference #	Method of measuring discharge	Rationale for sample locations	Source of discharge	Volume
03/28/2023 12:00 AM	Field 4	Kisst Dairy	Unknown	No sample was procured.	Storm water	1 *gal*

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period? Yes

Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? Yes

Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Kisst (May 14, 2024 12:59 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Kisst	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 14, 2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Discharge Maps
Provide map(s) showing the discharge and sample locations for each discharge or release of waste to land areas (land application areas or otherwise) or surface water.

Discharge Lab Reports
Provide copies of laboratory analyses of all discharges (manure, process wastewater, or tailwater), surface water (upstream and downstream of a discharge), and storm water, including chain-of-custody forms and laboratory quality assurance/quality control results.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Kisst Dairy
P.O. Box 300
Ripon, CA 95366

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well	T3079002A	Electrical Conductivity (EC)	EPA 120.1 and	4.30	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.535	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	2750	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: KISST DAIRY
Mailing address:
(not facility address) RIPON CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/17/23	9:20	DOMESTIC WELL	DOMESTIC	3.6 C		X				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-20-23 7:30			T3079002

Required to be submitted to any agency? (YES / NO)

FORM FOR
PRIORITY REPORTING OF SIGNIFICANT EVENTS
UNDER
WASTE DISCHARGE REQUIREMENTS GENERAL ORDER NO. R5-2007-0035
FOR
EXISTING MILK COW DAIRIES

A **significant event** is any noncompliance with Waste Discharge Requirements General Order No. R5-2007-0035 (General Order) that endangers human health or the environment, or any noncompliance with Prohibitions A1, A2, A3, A4, A5, A8, A9, A10, A11, or A12 of the General Order. **Discharge** means the release of waste, such as solid manure or process wastewater, to land, surface water, or groundwater. Most noncompliance will be from off- property discharge of solid manure or process wastewater into surface water, or on-property discharge of solid manure or process wastewater in a way not allowed under the General Order, such as the flooding of cropland with process wastewater.

Noncompliance does not have to include an off-property or improper on-property discharge of solid manure or process wastewater, but can include actions that are not allowed under the General Order, such as accepting non-dairy waste for disposal at a dairy.

A written report shall be submitted to the Central Valley Regional Water Quality Control Board (Central Valley Water Board) office **within two weeks** of the Discharger becoming aware of the significant event. The report shall contain a description of the noncompliance, its causes, duration, and the actual or anticipated time for achieving compliance. The report shall include complete details of the steps that the Discharger has taken or intends to take, in order to prevent recurrence. All intentional or accidental spills shall be reported as required by this provision.

Properly completing the following form will satisfy the requirement for the written report.

Item on page MRP-11	Description of Item	Response
1.	Date and time the noncompliance started, and location of the noncompliance (a map may be used to show the location)	March 28, 2023
	Was there a discharge of solid manure or process wastewater to surface water as a result of the noncompliance?	Wastewater
	If the noncompliance included a discharge to surface water, what was the ultimate destination of the discharge and the flow path of the discharge to the nearest water body? (a map may be used to show the flow path)	Old River

Item on page MRP-11	Description of Item	Response
2.	A description of the noncompliance (what happened) and why it happened	Using a tractor hydraulic pump, dairy was pumping ponded water from Field 4 into a field drain associated with Field 10. Dairy owner, John Kisst, assumed the ponded water was clean runoff from the recent rains. The discharge water traveled out of the field drain and into a regional where it was picked up by a pump which lifted the water into Old River.
3. If there was a discharge, what was the:	Flow rate of the discharge?	Unknown
	Total volume of waste discharged?	Unknown
	How long did the discharge last?	Unknown
4. If there was a discharge, what was the:	Amount of rain (in inches) the day of the discharge	.53 inches

If there was a discharge, complete the following table for the amount of rain on each of the 7 days before the discharge:

	Day before the discharge	2 days before the discharge	3 days before the discharge	4 days before the discharge	5 days before the discharge	6 days before the discharge	7 days before the discharge
Date	3/27/23	3/26/23	3/25/23	3/24/23	3/23/23	3/22/23	3/21/23
Rainfall (inches)	.00	.00	.00	.00	.00	.18	.28

Item on page MRP-11	Description of Item	Response
5. If there was a discharge:	How many samples were collected of the noncompliance discharge and/or surface water?	

Complete the attached Table (Summary of Information on Samples) for the samples you collected.

Note: Copies of the results of the laboratory analyses that you receive from the laboratory shall be submitted to the Central Valley Water Board office within 45 days of the discharge.

Item on page MRP-11	Description of Item	Response
6.	How long did the noncompliance last? (give dates and times) Note that the total period of noncompliance can be longer than any associated discharge, depending on what happened.	Unknown
	Has the noncompliance been corrected as of the date of this report?	Yes
	If the noncompliance has not been corrected, how much longer will it continue?	N/A

Attach a plan for actions that you will take to prevent this problem from happening in the future. <u>Include a time schedule for implementing the plan.</u>

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Kisst

Signature of owner or operator of the facility

John Kisst

Print or type name

Owner /Operator 4-30-23

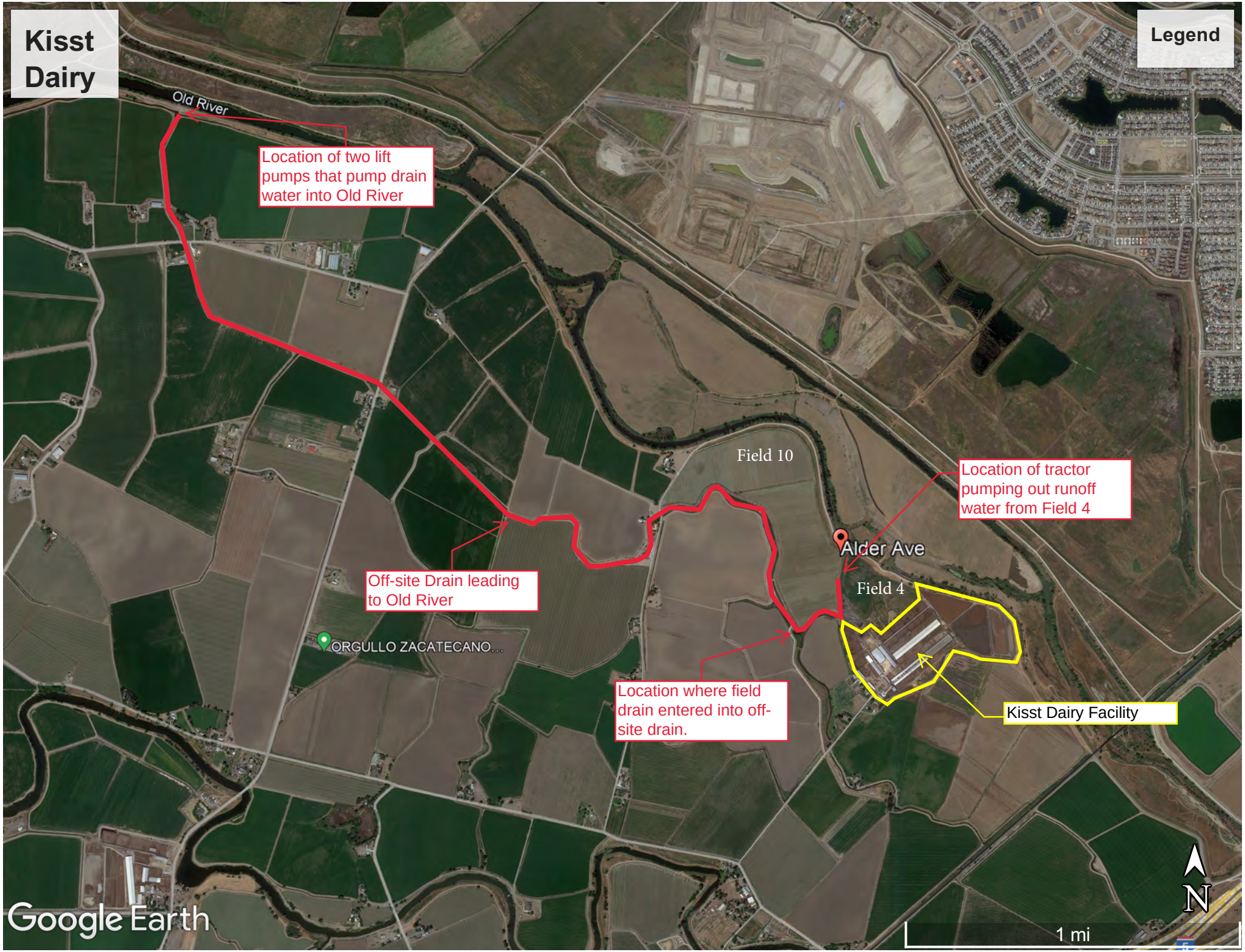
Title and Date

SUMMARY OF INFORMATION ON SAMPLES

Dairy Name:

Dairy Address:

Sample Identification Number	Location Where Collected	Date Collected	Time Collected	Field pH	Field Temperature	Field Dissolved Oxygen	Field EC	Date Sent to Laboratory	Analyses Requested


Kisst Dairy
Legend
Old River
Location of two lift pumps that pump drain water into Old River
Field 10
Location of tractor pumping out runoff water from Field 4
Alder Ave
Field 4
Off-site Drain leading to Old River
ORGULLO ZACATECANO...
Location where field drain entered into off-site drain.
Kisst Dairy Facility
N
1 mi
Google Earth

California Irrigation Management Information System (CIMIS)

CIMIS Daily Report

Rendered in ENGLISH Units.
Wednesday, March 1, 2023 - Friday, March 31, 2023
Printed on Friday, April 21, 2023

Holt - San Joaquin Valley - Station 248

Date	ETo (in)	Precip (in)	Sol Rad (Ly/day)	Avg Vap Pres (mBars)	Max Air Temp (°F)	Min Air Temp (°F)	Avg Air Temp (°F)	Max Rel Hum (%)	Min Rel Hum (%)	Avg Rel Hum (%)	Dew Point (°F)	Avg Wind Speed (mph)	Wind Run (miles)	Avg Soil Temp (°F)
3/1/2023	0.14	0.00	454 R	6.6	54.1	35.9	44.9	99	41	65	33.9	12.1 Y	290.2 Y	45.6
3/2/2023	0.10	0.00	443	6.7	61.4	30.5	46.6	99	29	62	34.2	4.5	108.9	46.6
3/3/2023	0.10	0.00	441	10.2	61.0	33.5	47.6	100	61	91	44.9	4.8	116.0	48.5
3/4/2023	0.07	0.03	287	8.8	55.4	39.2	46.7	98	57	82	41.3	7.9	188.8	48.9
3/5/2023	0.10	0.25	385	8.2	56.0	36.3	45.9	99	46	78	39.5	7.6	182.2	48.2
3/6/2023	0.09	0.00	396	8.5	56.9	33.3	44.4	100	52	85	40.2	4.0	95.5	48.4
3/7/2023	0.10	0.00	417	8.4	56.0	38.1	46.7	99	46	78	40.0	4.0	95.9	49.7
3/8/2023	0.11	0.06	452	9.0	59.5	38.4	47.2	100	46	81	41.7	4.6	110.6	50.9
3/9/2023	0.02	0.96	127	10.5	53.0	37.4	46.2	100	80	98	45.7	11.0 Y	264.0 Y	48.9
3/10/2023	0.04	0.46	199	13.0	59.9	50.4	53.3	100	75	94	51.5	7.5	179.3	51.5
3/11/2023	0.06	0.03	287	14.5	63.2	49.7	54.8	99	92	99	54.6	4.0	95.3	53.8
3/12/2023	0.03	0.45	182	15.0	61.3	48.4	55.5	100	99	100	55.4	5.3	127.2	54.6
3/13/2023	0.11	0.03	397	15.3	70.3	50.0	59.8	100	53	87	56.1	4.7	111.9	56.6
3/14/2023	0.04	0.26	179	15.2	67.2	46.7	56.9	100	66	96	55.8	12.8 R	306.4 R	56.7
3/15/2023	0.10	0.00	382	10.6	63.5	37.4	50.0	100	51	86	46.1	5.4	128.6	54.1
3/16/2023	0.13	0.00	513 R	10.0	65.2	40.0	51.7	100	49	77	44.6	4.0	95.6	54.4
3/17/2023	0.13	0.00	488	11.5	68.8	36.8	52.4	100	54	85	48.2	3.9	94.4	54.6
3/18/2023	0.09	0.00	343	12.8	67.5	39.3	55.3	100	55	86	51.1	4.0	95.6	54.8
3/19/2023	0.04	0.21	158	13.8	65.0	51.3	55.5	99	49	92	53.1	10.5 Y	252.7 Y	55.1
3/20/2023	0.11	0.00	419	11.7	61.0	45.4	52.9	100	64	85	48.6	4.7	113.2	56.1
3/21/2023	0.05	0.28	158	11.3	60.8	46.1	50.6	99	67	90	47.7	9.6	230.5	54.3
3/22/2023	0.07	0.18	312	12.2	58.0	47.4	51.5	99	76	94	49.9	9.2	221.1	53.3
3/23/2023	0.13	0.00	452	10.6	60.0	44.7	51.8	99	53	81	46.1	7.7	185.5	54.6
3/24/2023	0.15	0.00	552 R	7.8	59.9	38.6	49.0	98	38	66	38.2	7.5	180.4	53.7
3/25/2023	0.17	0.00	561 R	7.2	60.5	35.9	48.1	99	31	63	36.2	7.9	189.3	52.8
3/26/2023	0.15	0.00	565 R	6.9	59.5	33.9	46.4 Y	99	29	65 Y	35.1 Y	5.9	142.2	52.2
3/27/2023	0.15	0.00	545	8.1	62.7	31.7 Y	49.0	100	36	68	39.1	4.0	95.4	52.2
3/28/2023	0.03 R	0.53	80	9.3	55.8	38.8	47.3 Y	100	44	84 Y	42.6 Y	6.5	155.8	51.9
3/29/2023	0.03	0.22	204	10.3	52.4 Y	39.6	45.3 Y	100	96	100 Y	45.2 Y	5.8	139.8	50.4 Y
3/30/2023	0.12	0.00	485	10.1	61.3	42.8	49.7	100	54	83	44.9	4.6	110.3	52.6
3/31/2023	0.14	0.00	539	10.4	64.7	42.0	51.3	99	53	80	45.5	3.4	81.9	54.6
Tots/Avgs	2.90	3.95	368	10.5	60.7	40.6	50.1	99	56	83	45.1	6.4	154.3	52.3

Flag Legend		
A - Historical Average	I - Ignore	R - Far out of normal range
C or N - Not Collected	M - Missing Data	S - Not in service
H - Hourly Missing or Flagged Data	Q - Related Sensor Missing	Y - Moderately out of range
Conversion Factors		
Ly/day/2.065=W/sq.m	inches * 25.4 = mm	(F-32) * 5/9 = c
mph * 0.447 = m/s	mBars * 0.1 = kPa	miles * 1.60934 = km

2023 Kisst Dairy Annual Report

Final Audit Report 2024-05-14

Created:	2024-05-14
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA3PxQle6vxWox4dR6Fjr36liYEl59Ndyu

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